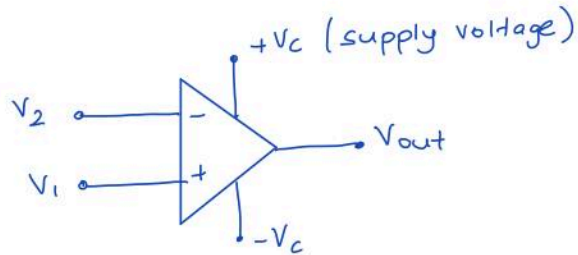
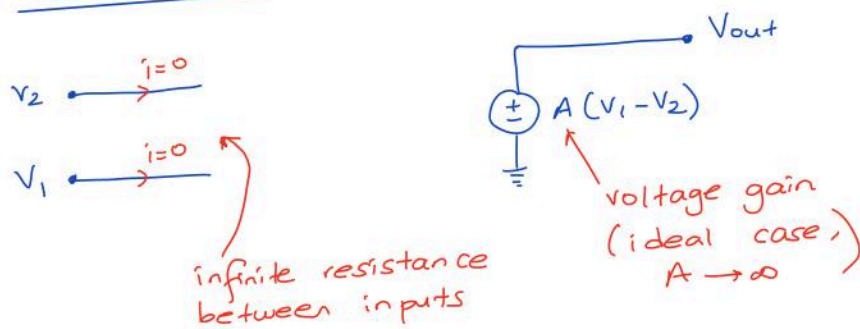


Operational Amplifier (OPAMP) :

- used to amplify a signal, and/or perform operations on a signal



Ideal OPAMP model :



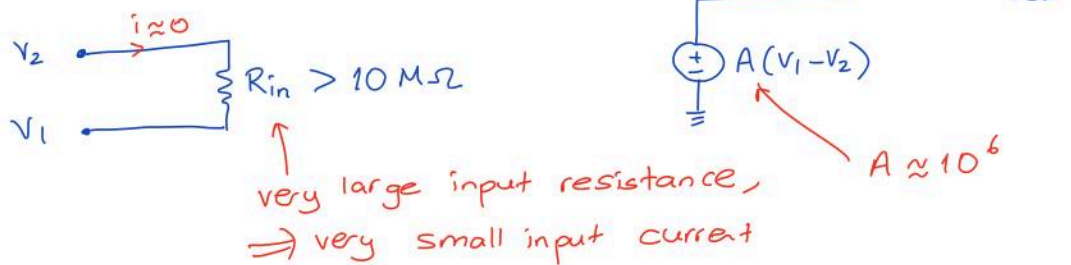
$$V_{out} = A(V_1 - V_2)$$

So, as long as V_{out} is finite :

$$V_1 - V_2 = \frac{V_{out}}{A} \rightarrow 0$$

$$\boxed{V_1 = V_2}$$

Real OPAMP model :



$$V_{out} \approx A(V_1 - V_2)$$

So,

$$V_{out} \approx A(V_1 - V_2)$$

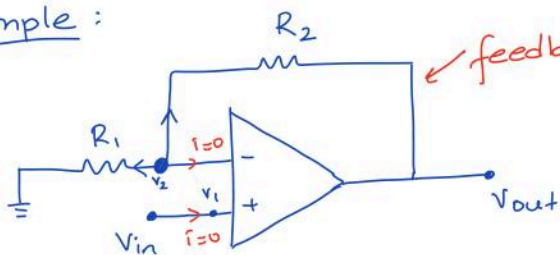
$$V_1 - V_2 \approx \frac{V_{out}}{A} \quad \begin{matrix} \leftarrow \text{in } V_{out} \text{ range} \\ \leftarrow 10^6 \end{matrix}$$

So,

$$\boxed{V_1 \approx V_2}$$

* In this class, we will use the ideal OPAMP model.

Example :



feedback loop * $i = 0$ into the OPAMP
* $V_1 = V_2$

$$V_1 = V_{in} = V_2$$

* KCL at V_2 :
$$\frac{V_2}{R_1} + \frac{(V_2 - V_{out})}{R_2} = 0$$

$$\frac{V_{in}}{R_1} + \frac{V_{in} - V_{out}}{R_2} = 0$$

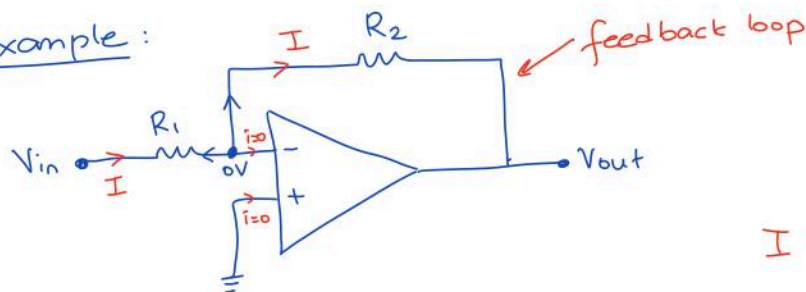
$$(R_1 + R_2) V_{in} = R_1 V_{out}$$

$$V_{out} = \left(1 + \frac{R_2}{R_1}\right) V_{in}$$

gain of the OPAMP circuit

* Called a non-inverting amplifier. Gain is positive valued.

Example :



$$I = \frac{V_{in}}{R_1} = -\frac{V_{out}}{R_2}$$

$$V_+ = 0$$

$$V_+ = V_- = 0$$

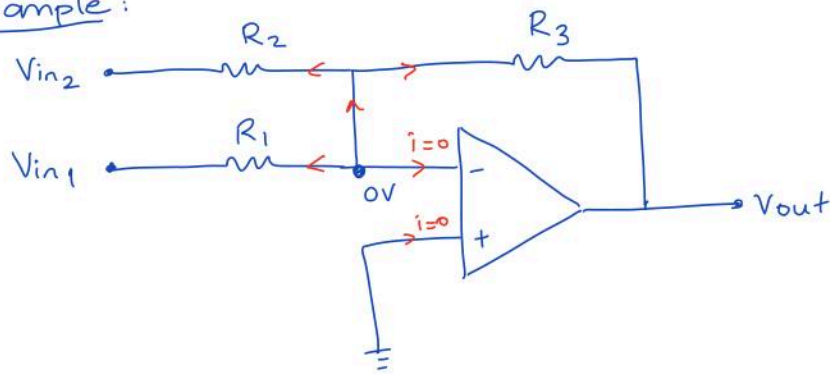
KCL :
$$\frac{0 - V_{in}}{R_1} + \frac{0 - V_{out}}{R_2} = 0$$

$$V_{out} = -\frac{R_2}{R_1} V_{in}$$

overall gain of OPAMP circuit

* Called an inverting amplifier (due to negative overall gain)

Example:



$$V_+ = V_- = 0$$

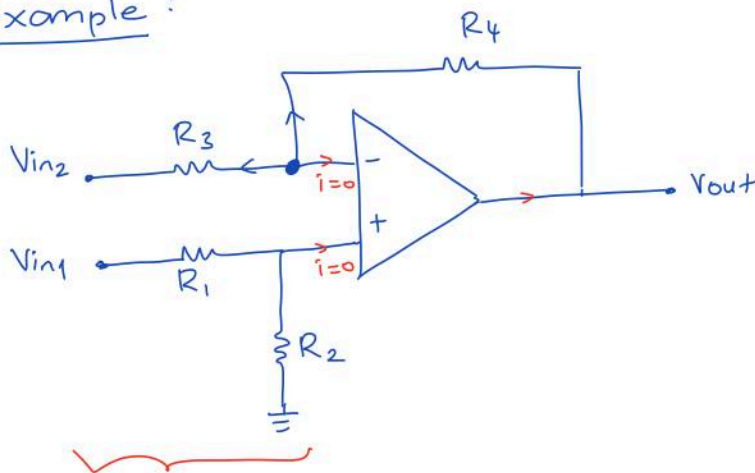
KCL :
$$\frac{0 - V_{in1}}{R_1} + \frac{0 - V_{in2}}{R_2} + \frac{0 - V_{out}}{R_3} = 0$$

$$V_{out} = -\frac{R_3}{R_1} V_{in1} - \frac{R_3}{R_2} V_{in2}$$

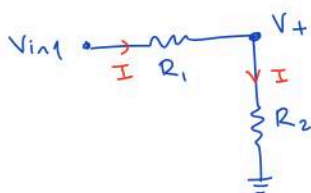
$$= - \left(\frac{R_3}{R_1} V_{in1} + \frac{R_3}{R_2} V_{in2} \right)$$

summing amplifier

Example:



$$V_+ = V_-$$



$$I = \frac{V_{in1}}{R_1 + R_2}$$

$$V_+ = I \cdot R_2 = \frac{R_2}{R_1 + R_2} V_{in1}$$

* KCL :
$$\frac{V_- - V_{in2}}{R_3} + \frac{V_- - V_{out}}{R_4} = 0$$

$$(R_3 + R_4) V_- - R_4 V_{in2} = R_3 V_{out}$$

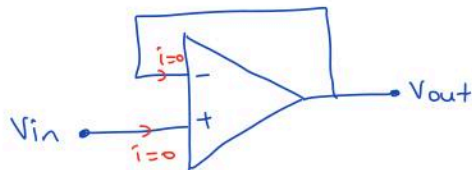
$$(R_3 + R_4) \frac{R_2}{R_1 + R_2} V_{in1} - R_4 V_{in2} = R_3 V_{out}$$

$$V_+ = V_-$$

$$V_{out} = \frac{R_3 + R_4}{R_3} \cdot \frac{R_2}{R_1 + R_2} V_{in1} - \frac{R_4}{R_3} V_{in2}$$

difference amplifier

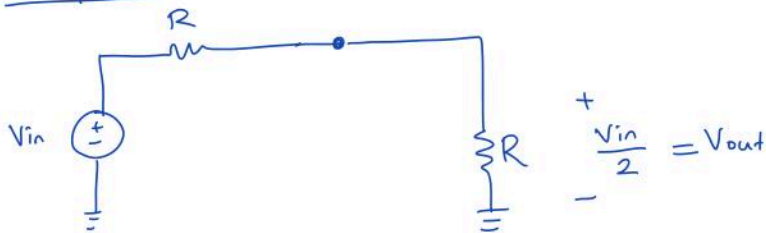
Example :



$$\begin{aligned} V_+ &= V_{in} \\ V_- &= V_{out} \\ V_+ &= V_- \Rightarrow V_{out} = V_{in} \end{aligned}$$

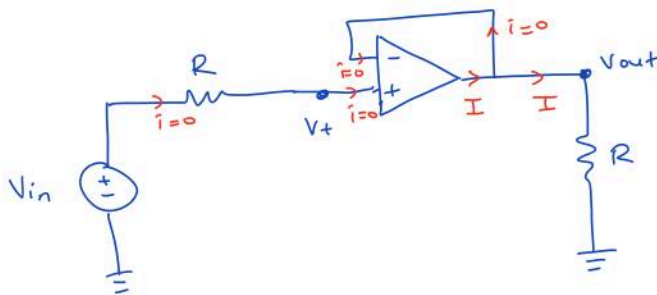
* This is a buffer.
Transfers the input voltage intact to the output.

Compare :



$$P = \frac{V^2}{R} = \frac{V_{in}^2}{4R}$$

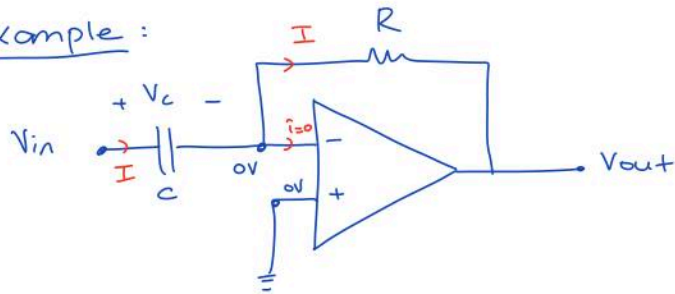
VS.



$$\begin{aligned} V_+ &= V_{in} \\ V_- &= V_{out} \\ V_{out} &= V_{in} \end{aligned}$$

$$P = \frac{V_{in}^2}{R}$$

Drives load as a perfect voltage source of $V = V_{in}$.

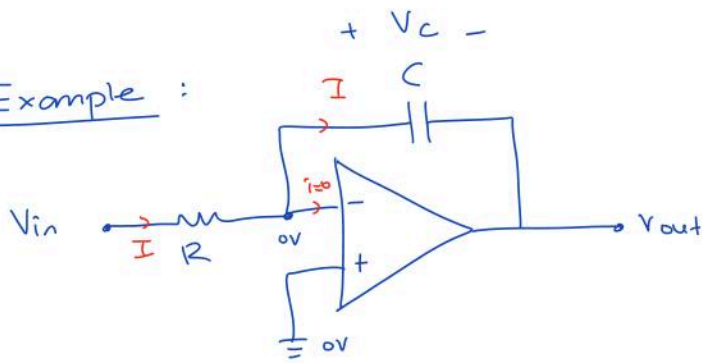
Example : V_{in} has arbitrary shape

$$\left. \begin{aligned} I &= C \frac{dV_c}{dt} = C \frac{dV_{in}}{dt} \\ I &= -\frac{V_{out}}{R} \end{aligned} \right\}$$

$$-\frac{V_{out}}{R} = C \frac{dV_{in}}{dt}$$

$$V_{out} = -RC \frac{dV_{in}}{dt}$$

differentiator amplifier

Example :

$$I = \frac{V_{in} - 0}{R}$$

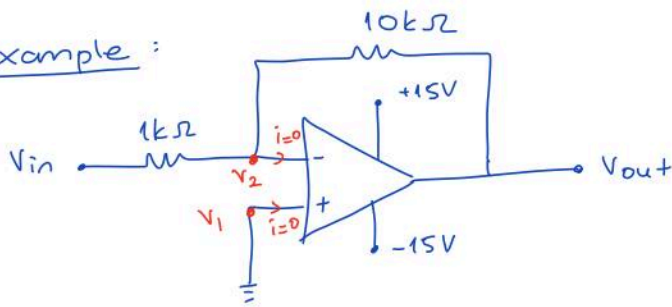
$$I = C \frac{dV_c}{dt} = -C \frac{dV_{out}}{dt}$$

$$\frac{V_{in}}{R} = -C \frac{dV_{out}}{dt}$$

$$V_{out} = -\frac{1}{RC} \int V_{in}(t) dt$$

integrator amplifier

Example :



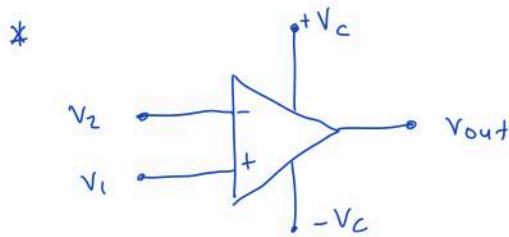
$$V_{out} = -\frac{10k\Omega}{1k\Omega} V_{in}$$

$$\frac{V_{out}}{V_{in}} = -10$$

Case 1 : $V_{in} = 1V$
 $V_{out} = -10V$ ✓ OPAMP can supply this

Case 2 : $V_{in} = 10V$
 $V_{out} = -100V$ ✗ No.

OPAMP output has to be in the ranges of supply voltages ($\pm V_c$)



V_{max} : max output of OPAMP
 $V_{max} \approx +V_c$

V_{min} : min output of OPAMP
 $V_{min} \approx -V_c$

OPAMP is in linear mode

So,

$$V_{out} = \begin{cases} A(V_1 - V_2) & \text{if } V_{min} \leq A(V_1 - V_2) \leq V_{max} \\ V_{max} & \text{if } A(V_1 - V_2) > V_{max} \\ V_{min} & \text{if } A(V_1 - V_2) < V_{min} \end{cases}$$

OPAMP is saturated

* During saturation, $V_1 \approx V_2$ is no longer valid!
 ($I=0$ into the OPAMP is still valid)

* Back to example :

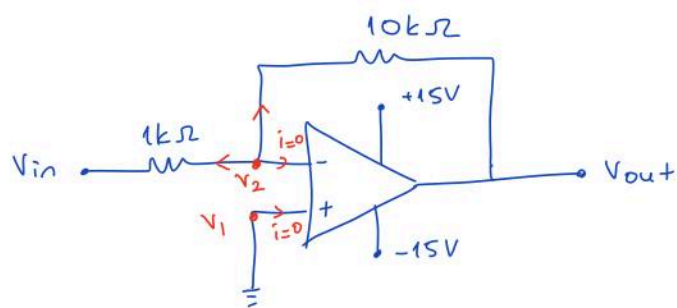
if $V_{in} = 10V$
 $V_{out} = -100V$

✗ OPAMP cannot provide this output

Then, $V_{out} = V_{min}$ (saturated)

Assuming $V_{min} \approx -V_c = -15V$

$$\Rightarrow V_{out} = -15V$$



Now, calculate V_2 :

$$\text{KCL: } \frac{V_2 - V_{in}}{1k} + \frac{V_2 - V_{out}}{10k} = 0$$

$$11 V_2 = 10 V_{in} + V_{out}$$

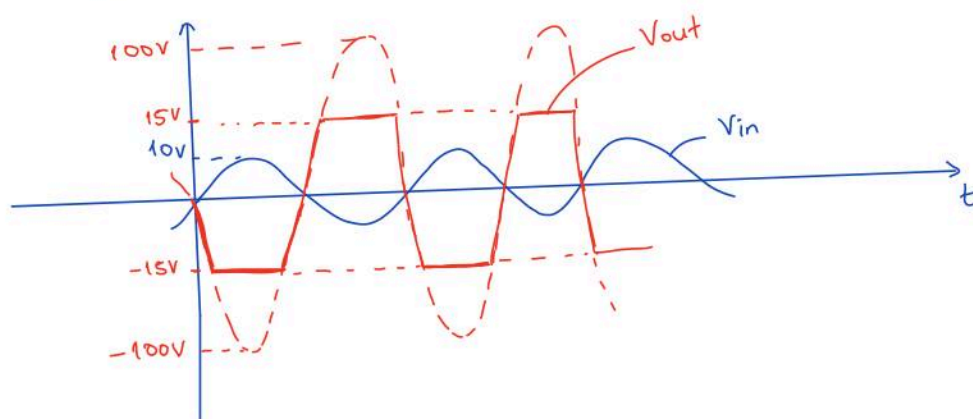
$$11 V_2 = 10 \cdot 10 - 15$$

$$V_2 = \frac{85}{11} \approx 7.7V$$

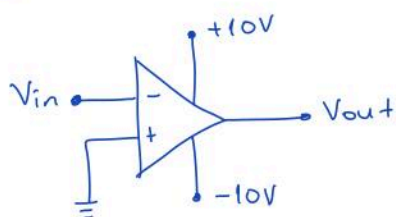
So, $V_1 \neq V_2$ ($V_1 = 0V$)

$(V_1 - V_2) = \underbrace{-7.7V}_{\text{negative difference}} \rightarrow \text{negative saturation.}$

* if $V_{in} = 10 \sin(2\pi f_0 t) V$



Another OPAMP circuit:



$$V_2 = V_{in}(t) = 2 \sin(2\pi f_0 t) V$$

$$V_1 = 0V$$

$$V_1 \neq V_2$$

$\Rightarrow$ OPAMP is always saturated.

$$V_{out}(t) = A (V_1 - V_2)$$

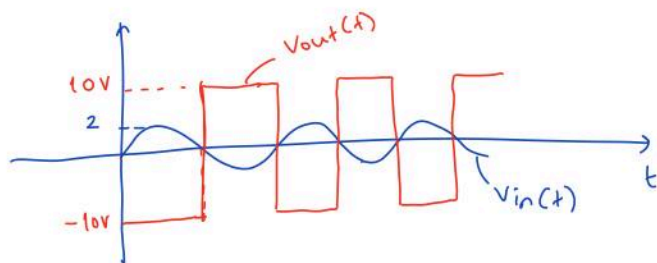
$$= -A \cdot V_2(t)$$

$$= -A \cdot 2 \cdot \sin(2\pi f_0 t)$$

$$\uparrow$$

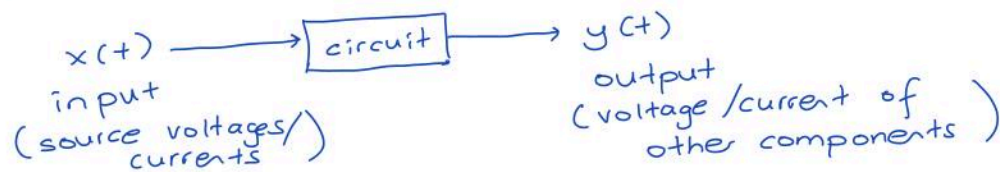
$$\infty$$

$V_{out}(t)$ is a square wave.



Linear Circuits: Any circuit made up of resistors, capacitors, inductors, OPAMPs (if not saturated) is linear.

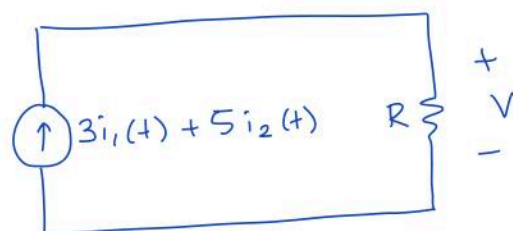
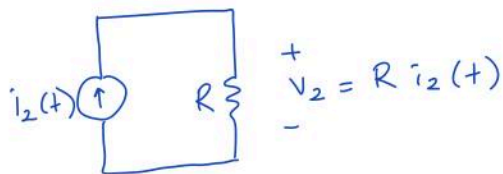
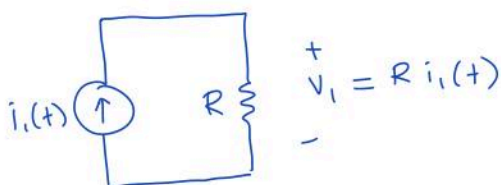
Linearity:



$$\text{if } \begin{array}{l} x_1(t) \longrightarrow y_1(t) \\ x_2(t) \longrightarrow y_2(t) \end{array}$$

$$\text{then, } a \cdot x_1(t) + b \cdot x_2(t) \longrightarrow a y_1(t) + b y_2(t)$$

Example: resistor



$$\begin{aligned} V &= [3i_1(t) + 5i_2(t)] \cdot R \\ &= 3 \cdot R i_1(t) + 5 R i_2(t) \\ &= 3 \cdot V_1(t) + 5 V_2(t) \end{aligned}$$

* So, resistors are linear circuit elements.

* Same can be shown for capacitors and inductors.

For linear circuits:

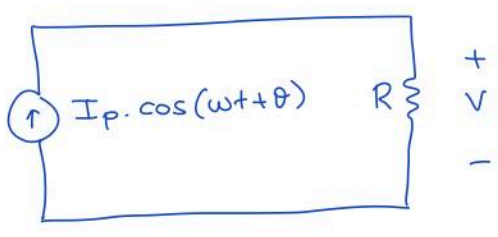
If input is a sinusoidal voltage or current, the output (any voltage or current in the circuit) is also sinusoidal at the same frequency.

$$\text{Input} = A \cdot \sin(\omega t + \theta_1)$$

$$\text{Output} = B \cdot \sin(\omega t + \theta_2)$$

} some frequency.
Amplitudes & phases may be different.

Resistors :



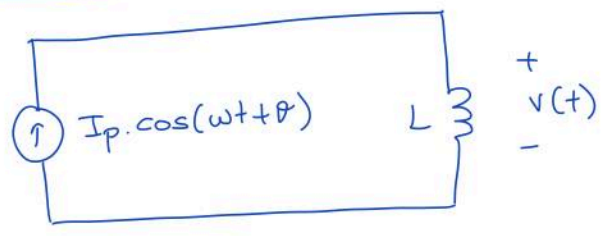
$$v(t) = R \cdot I_p \cdot \cos(\omega t + \theta)$$

$$= V_p \cdot \cos(\omega t + \theta)$$

amplitude depends on resistance

same freq.
same phase.

Inductor :



$$v(t) = L \frac{di}{dt}$$

$$= L \frac{d}{dt} (I_p \cdot \cos(\omega t + \theta))$$

$$= -\omega I_p L \sin(\omega t + \theta)$$

$$= \omega L I_p \cdot \cos(\omega t + \theta + \frac{\pi}{2})$$

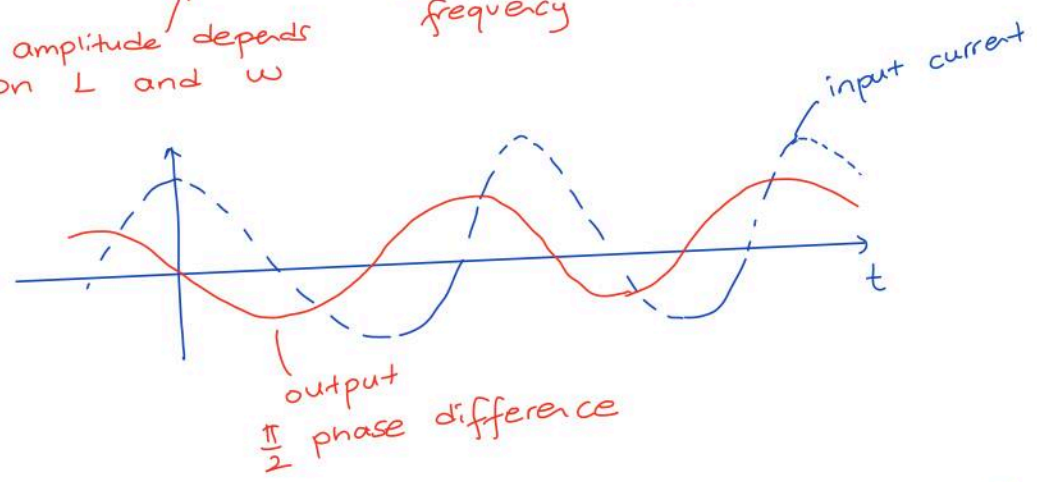
using $-\sin(x) = \cos(x + \frac{\pi}{2})$

$$v(t) = \underbrace{\omega L I_p}_{V_p} \cdot \cos(\omega t + \theta + \frac{\pi}{2})$$

amplitude depends on L and ω

same frequency

$\frac{\pi}{2}$ phase shift



*** Linear circuits do not change the input frequency.
They may change the amplitude and phase only.

Complex Numbers - Review

(10)

$$z = a + jb, \quad j = \sqrt{-1}$$

$$\operatorname{Re}\{z\} = a$$

$$\operatorname{Im}\{z\} = b$$

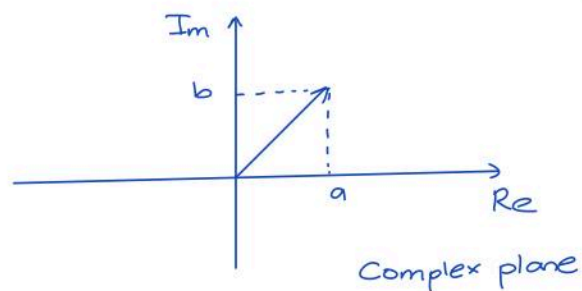
$$z^* = a - jb$$

$$|z| = |a + jb| = \sqrt{a^2 + b^2}$$

$$z \cdot z^* = a^2 + b^2$$

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad \text{if } z_2 \neq 0$$



* Relation between sinusoids and complex exponential functions (Euler's formula):

$$e^{j\theta} = \cos \theta + j \sin \theta$$

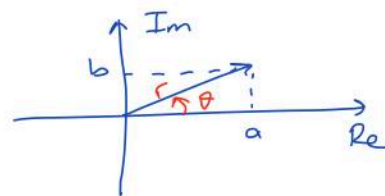
$$\cos \theta = \operatorname{Re}\{e^{j\theta}\} = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin \theta = \operatorname{Im}\{e^{j\theta}\} = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$|e^{j\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1, \quad \text{regardless of } \theta$$

* Complex numbers in polar coordinates:

$$z = a + jb \Rightarrow \begin{aligned} r &= \sqrt{a^2 + b^2} \\ \theta &= \tan^{-1}\left(\frac{b}{a}\right) \end{aligned}$$



$$z = r \cdot e^{j\theta} = r \angle \underset{\substack{\uparrow \\ \text{phase}}}{\theta} = r (\cos \theta + j \sin \theta)$$

$$a = r \cos \theta$$

$$b = r \sin \theta$$

$$* \text{ so, } z_1 \cdot z_2 = r_1 \cdot e^{j\theta_1} \cdot r_2 \cdot e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$$

Phasor Notation

(11)

Phasors represent sine waves of different amplitudes and phases, but all at the same frequency.

$$A \cdot \cos(\omega t + \theta)$$

in time domain

$$\Rightarrow A \cdot e^{j\theta}$$

amplitude $\rightarrow$ phase
in phasor domain

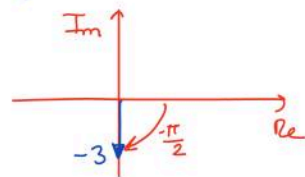
To convert to time domain:

$$\begin{aligned} & \operatorname{Re} \{ A \cdot e^{j\theta} \cdot e^{j\omega t} \} \\ &= \operatorname{Re} \{ A e^{j(\omega t + \theta)} \} \\ &= A \cos(\omega t + \theta) \end{aligned}$$

Example: $5 \cos(\omega t) \Rightarrow 5 \cdot e^{j0} = 5$

$$6 \cos(\omega t + 30^\circ) \Rightarrow 6 \cdot e^{j30^\circ} = 6 \cdot e^{j\frac{\pi}{6}}$$

$$3 \sin(\omega t) = 3 \cos(\omega t - \frac{\pi}{2}) \Rightarrow 3 \cdot e^{-j\pi/2} = -3j$$

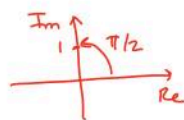


$$5 e^{-j20^\circ} \Rightarrow 5 \cos(\omega t - 20^\circ)$$

$$4 + 4j = 4\sqrt{2} e^{j\frac{\pi}{4}} \Rightarrow 4\sqrt{2} \cos(\omega t + \frac{\pi}{4})$$

Derivative: $A \cos(\omega t + \theta) \Rightarrow A e^{j\theta}$

$$\begin{aligned} \frac{d}{dt} A \cos(\omega t + \theta) &= -A\omega \sin(\omega t + \theta) \\ &= A\omega \cos(\omega t + \theta + \frac{\pi}{2}) \end{aligned}$$



$$\begin{aligned} & \Rightarrow A\omega e^{j(\theta + \frac{\pi}{2})} \\ &= A \cdot \omega \underbrace{e^{j\frac{\pi}{2}}}_j \cdot e^{j\theta} \\ &= j\omega A \cdot e^{j\theta} \end{aligned}$$

derivative in time domain $\Rightarrow$ multiplication with $j\omega$ in phasor domain

Integration :

$$A \cos(\omega t + \theta) \Rightarrow A e^{j\theta}$$

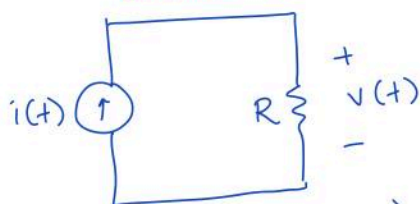
$$\int A \cos(\omega t + \theta) dt = \frac{A}{\omega} \sin(\omega t + \theta) = \frac{A}{\omega} \cos(\omega t + \theta - \frac{\pi}{2})$$

$$\begin{aligned} \Rightarrow \frac{A}{\omega} e^{j(\theta - \frac{\pi}{2})} &= \frac{A}{\omega} \cdot e^{j\theta} \cdot \underbrace{e^{-j\frac{\pi}{2}}}_{-j} \\ &= \frac{-j(j)}{\omega(j)} A e^{j\theta} \\ &= \frac{1}{j\omega} \cdot A e^{j\theta} \end{aligned}$$

integration in time domain $\Rightarrow$ division by $j\omega$ in phasor domain

Resistors :

in time domain



$$i(t) = I_p \cdot \cos(\omega t + \theta)$$

$$v(t) = i(t) \cdot R = R \cdot I_p \cdot \cos(\omega t + \theta)$$

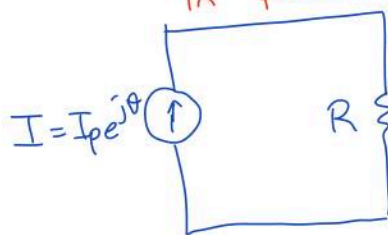
$$I = I_p \cdot e^{j\theta}$$

$$V = R \cdot \underbrace{I_p \cdot e^{j\theta}}_I$$

$$V = R \cdot I \quad \leftarrow \text{resistor has the same relation between its current and voltage in phasor domain.}$$

$$\frac{V}{I} = R$$

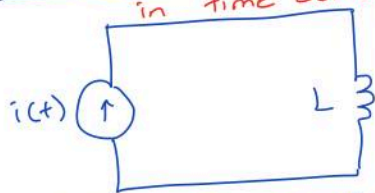
in phasor domain



$$V = R \cdot I = R I_p \cdot e^{j\theta}$$

Inductor :

in time domain



$$i(t) = I_p \cdot \cos(\omega t + \theta)$$

$$v(t) = \omega L I_p \cdot \cos(\omega t + \theta + \frac{\pi}{2}) \quad (\text{from before})$$

$$\Downarrow \\ I = I_p \cdot e^{j\theta}$$

$$v(t) = L \frac{di}{dt} \quad \text{convert to phasor domain}$$

$$V = L j\omega \cdot I$$

$$\boxed{V = j\omega L I}$$

$$= j\omega L I_p \cdot e^{j\theta}$$

$$= e^{j\pi/2} \cdot \omega L I_p e^{j\theta}$$

$$= \omega L I_p \cdot e^{j(\theta + \pi/2)}$$

↓ to time domain

$$v(t) = \omega L I_p \cos(\omega t + \theta + \frac{\pi}{2})$$

(same as before)

$$V = j\omega L I$$

$$\boxed{Z = \frac{V}{I} = j\omega L}$$

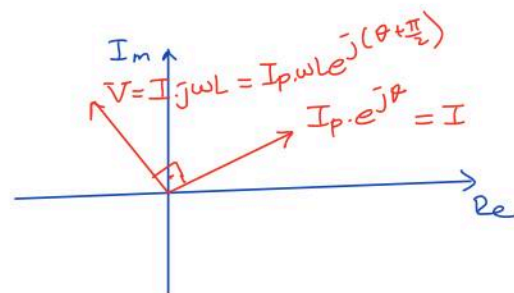
is called the impedance of the inductor.

in units of Ω

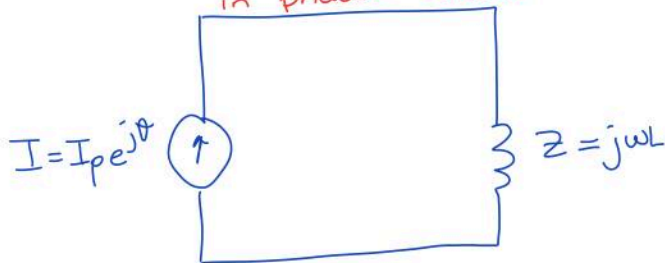
$$Z = j\omega L$$

amplitude ratio between voltage and current

$\frac{\pi}{2}$ phase difference between voltage and current



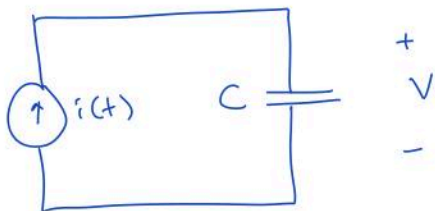
in phasor domain



+

$$V = Z \cdot I$$

-

Capacitor :

$$i(t) = C \frac{dv}{dt}$$

$$I = C \cdot j\omega V$$

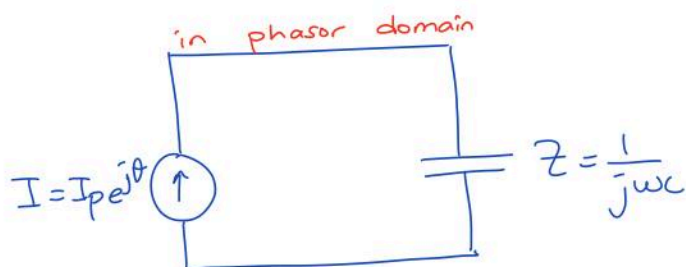
$$Z = \frac{V}{I} = \frac{1}{j\omega C}$$

impedance of capacitor

 $-\frac{\pi}{2}$ phase difference

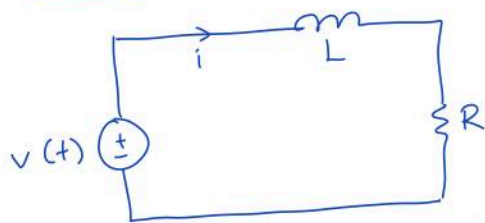
$$V = \frac{I}{j\omega C} = \frac{I_p \cdot e^{j\theta}}{j\omega C} = \frac{I_p \cdot e^{j\theta}}{e^{j\pi/2} \cdot \omega C} = \frac{I_p}{\omega C} e^{j(\theta - \frac{\pi}{2})}$$

$$\Rightarrow v(t) = \frac{I_p}{\omega C} \cos(\omega t + \theta - \frac{\pi}{2})$$



$$V = I \cdot Z$$

Impedance : like an extension of the concept of resistance in phasor domain. It expresses both the amplitude and phase.

Example :

$$v(t) = V_p \cdot \cos(\omega t + \theta)$$

$$v(t) = L \frac{di}{dt} + iR \quad (\text{in time domain})$$

$$\Downarrow$$

$$V = L \cdot j\omega I + I \cdot R \quad (\text{in phasor domain})$$

$$V = (j\omega L + R) \cdot I$$

$$Z = \frac{V}{I} = \underbrace{j\omega L}_{Z_L} + \underbrace{R}_{Z_R} \quad (\text{in } \Omega)$$

total impedance

In general : $Z = R + jX$ (impedance) (in Ω)

$$\text{Re}\{Z\} = R = \text{resistance}$$

$$\text{Im}\{Z\} = X = \text{reactance}$$

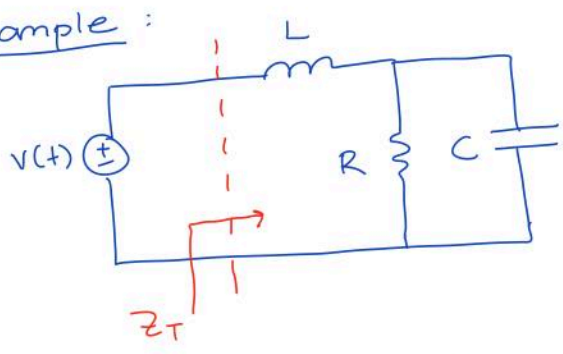
$$Y = \frac{1}{Z} = G + jB \quad (\text{admittance}) \quad (\text{in } S)$$

$$\text{Re}\{Y\} = G = \text{conductance}$$

$$\text{Im}\{Y\} = B = \text{susceptance}$$

Series/Parallel connection : works the same as in the case of series/parallel connection of resistors.

Example :



Z_T : total impedance

$$Z_T = (R // Z_C) + Z_L$$

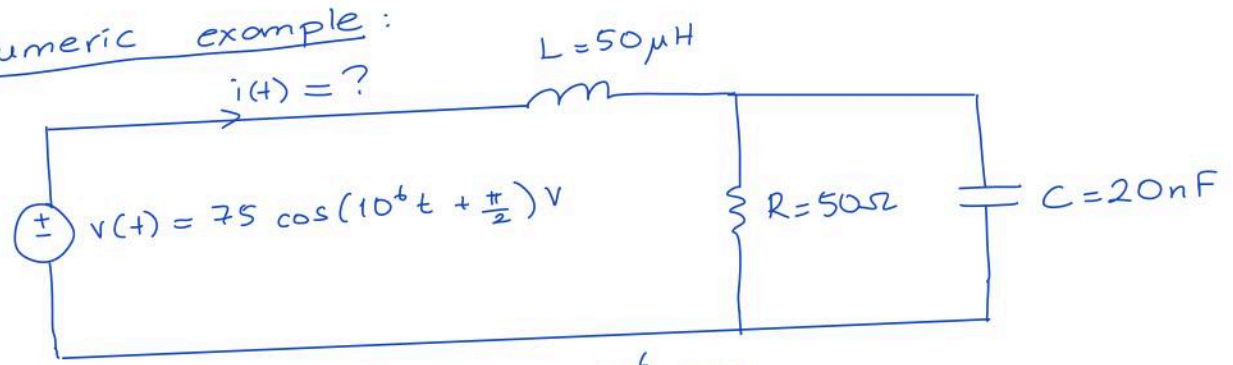
where $Z_C = \frac{1}{j\omega C}$

$$Z_L = j\omega L$$

$$Z_T = \left(R // \frac{1}{j\omega C} \right) + j\omega L$$

$$= \left(\frac{1}{R} + j\omega C \right)^{-1} + j\omega L = \frac{R}{1 + j\omega RC} + j\omega L$$

Numeric example :



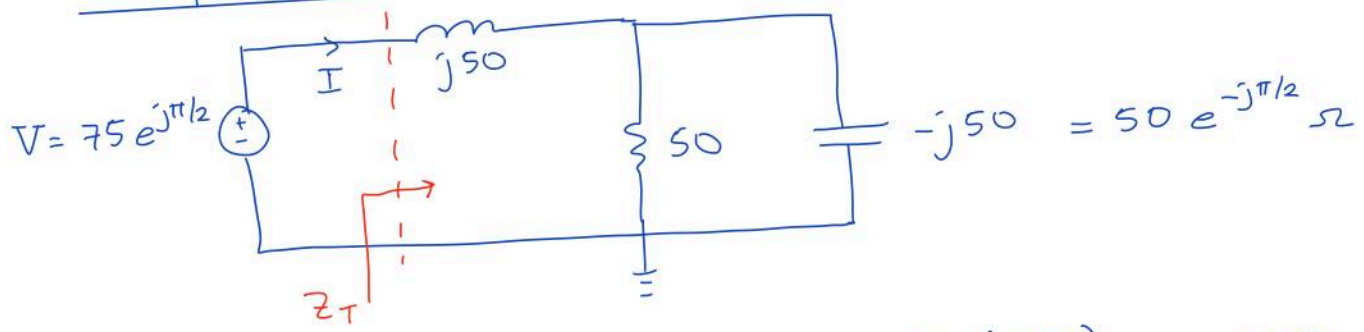
What is ω ? $\omega = 10^6 \text{ rps}$

$$Z_L = j\omega L = j 10^6 \cdot 50 \cdot 10^{-6} = j 50 \Omega$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j 10^6 \cdot 20 \cdot 10^{-9}} = \frac{1000 (-j)}{j 20 (-j)} = -j 50 \Omega$$

$$V = 75 e^{j\frac{\pi}{2}} \text{ V}$$

In phasor domain :



$$Z_T = (50 \parallel -j50) + j50 = \frac{50 \cdot (-j50)}{50 - j50} + j50$$

$$= \frac{-j50(1+j)}{1-j(1+j)} + j50 = \frac{-j50 + 50}{2} + j50$$

$$= -j25 + 25 + j50 = \boxed{25 + j25 \Omega}$$

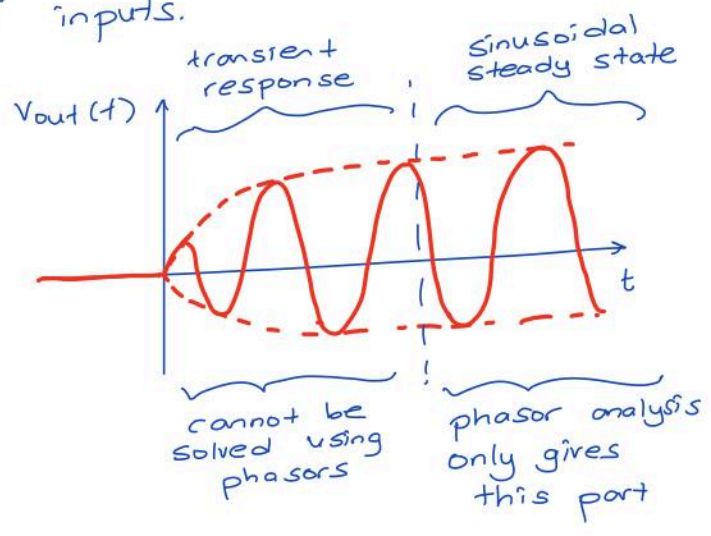
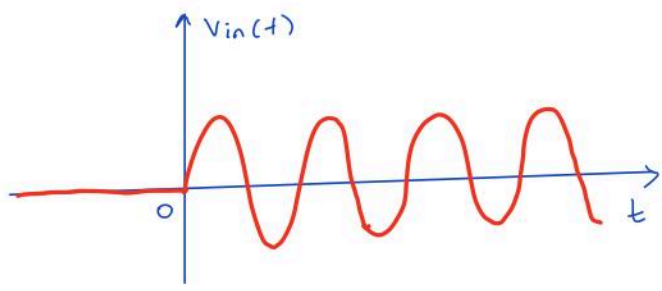
$$= 25\sqrt{2} e^{j\pi/4} \Omega$$

$$I = \frac{V}{Z_T} = \frac{75 e^{j\pi/2}}{25\sqrt{2} e^{j\pi/4}} = \frac{3}{\sqrt{2}} e^{j\pi/4} A$$

Finally, convert back to time domain:

$$i(t) = \frac{3}{\sqrt{2}} \cos(10^6 t + \frac{\pi}{4}) A$$

*** Use phasor analysis only for sinusoidal inputs. We have a sinusoidal input, and we assume that it has been on for a long time. So, the circuit has reached a steady state. Phasors are used to find this steady state solution for sinusoidal inputs.

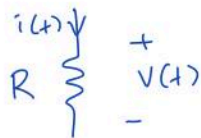


Last time:

Phasor notation:

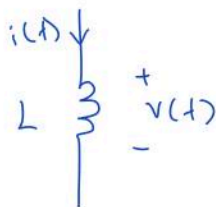
Time Domain	Phasor Domain
$A \cos(\omega t + \phi)$	$A e^{j\phi}$
Derivative	multiplication with $j\omega$
Integral	division by $j\omega$

Resistor:



$$Z = \frac{V}{I} = R$$

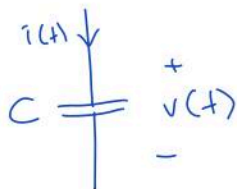
Inductor:



$$Z = \frac{V}{I} = j\omega L = \omega L e^{j\pi/2}$$

Note: at $\omega = 0$ (DC), $Z = 0 \Rightarrow$ short circuit
as $\omega \rightarrow \infty$, $Z \rightarrow \infty \Rightarrow$ does not allow rapid changes in current flow

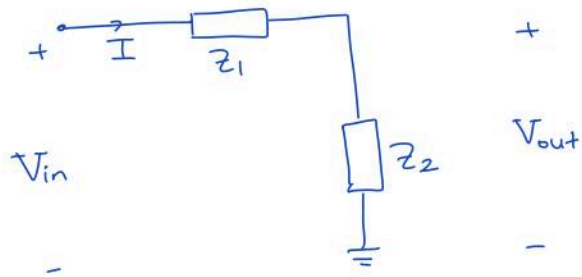
Capacitor:



$$Z = \frac{V}{I} = \frac{1}{j\omega C} = \frac{1}{\omega C} e^{-j\pi/2}$$

Note: at $\omega = 0$ (DC), $Z \rightarrow \infty \Rightarrow$ open circuit
as $\omega \rightarrow \infty$, $Z \rightarrow 0 \Rightarrow$ does not allow rapid changes in voltage

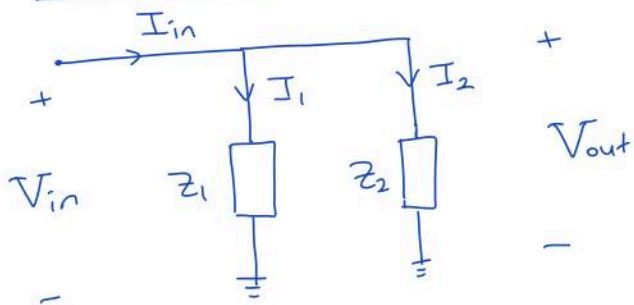
Voltage Divider: Same as in resistors



$$I = \frac{V_{in}}{Z_1 + Z_2}$$

$$V_{out} = I \cdot Z_2 = V_{in} \cdot \underbrace{\frac{Z_2}{Z_1 + Z_2}}_{\text{voltage divider term}}$$

Current Divider: Same as in resistors



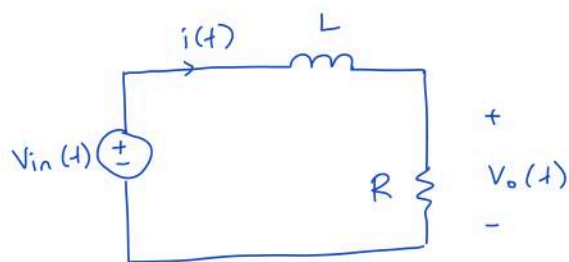
$$I_{in} = \frac{V_{out}}{Z_1 // Z_2}$$

$$I_2 = \frac{V_{out}}{Z_2}$$

$$I_2 = \frac{I_{in} \cdot (Z_1 // Z_2)}{Z_2} = \frac{I_{in} \cdot \frac{Z_1 \cdot Z_2}{Z_1 + Z_2}}{Z_2}$$

$$I_2 = I_{in} \cdot \underbrace{\frac{Z_1}{Z_1 + Z_2}}_{\text{current divider term}}$$

Transfer Function : Denotes how input voltage (or current) is scaled at the output



$$V_o = V_{in} \cdot \frac{R}{R + j\omega L} \quad (\text{voltage divider})$$

Transfer Function : $H(\omega) = \frac{V_o}{V_{in}} = \frac{R}{R + j\omega L} = \frac{1}{1 + j\omega \frac{L}{R}}$

ratio depends on frequency ω (pointing to $H(\omega)$)

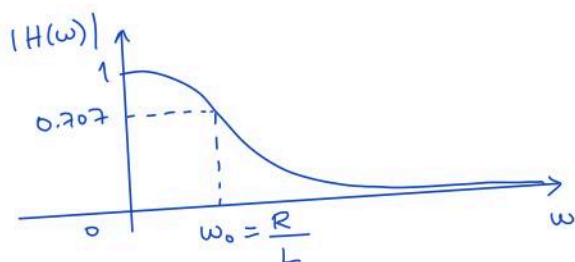
output phasor (pointing to V_o)

input phasor (pointing to V_{in})

Amplitude : $|H(\omega)| = \frac{1}{\sqrt{1 + \frac{\omega^2 L^2}{R^2}}}$

* at $\omega = 0$, $|H(\omega)| = 1 = |H(\omega)|_{\max}$

* as $\omega \rightarrow \infty$, $|H(\omega)| \rightarrow 0$



called a Low Pass Filter (LPF)

* at low frequencies, output amplitude is approximately the same as input amplitude

* output reaches zero as frequency increases.

* Special case : when $\omega_0 L = R$ OR $\omega_0 = \frac{R}{L}$

$$H(\omega) = \frac{1}{1 + j}$$

$$|H(\omega)| = \frac{1}{\sqrt{2}} \approx 0.707$$

$$= \frac{|H(\omega)|_{\max}}{\sqrt{2}}$$

ω_0 : cut-off frequency

Decibel : $dB \equiv 20 \log_{10}$ (Ratio of voltage amplitudes)

at $\omega_0 = \frac{R}{L}$, $|H(\omega_0)| = \frac{1}{\sqrt{2}}$

$$dB = 20 \cdot \log_{10} \left(\frac{1}{\sqrt{2}} \right) = -20 \log_{10} \sqrt{2} \approx -3 \text{ dB}$$

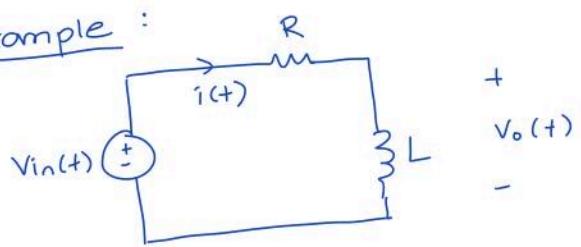
* $\omega_0 = \frac{R}{L}$ is called the 3-dB cut-off frequency

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \cdot \frac{R}{L} = \frac{1}{2\pi \cdot \frac{L}{R}}$$

* $|H(\omega=0)| = 1 \Rightarrow 0 \text{ dB}$

So, there is 3dB attenuation at the output at $\omega_0 = \frac{R}{L}$:

Example :



$$I = \frac{V_{in}}{R + j\omega L}$$

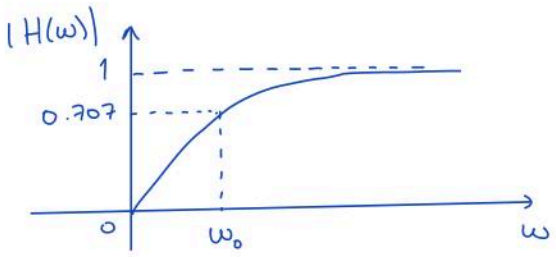
$$V_o = I \cdot j\omega L = V_{in} \cdot \frac{j\omega L}{R + j\omega L}$$

$$H(\omega) = \frac{V_o}{V_{in}} = \frac{j\omega L}{R + j\omega L} = \frac{j\omega L/R}{1 + j\omega L/R}$$

$$|H(\omega)| = \frac{\omega L/R}{\sqrt{1 + \frac{\omega^2 L^2}{R^2}}}$$

* At $\omega=0$, $|H(\omega)| = 0$

* As $\omega \rightarrow \infty$, $|H(\omega)| \rightarrow 1 = |H(\omega)|_{\max}$



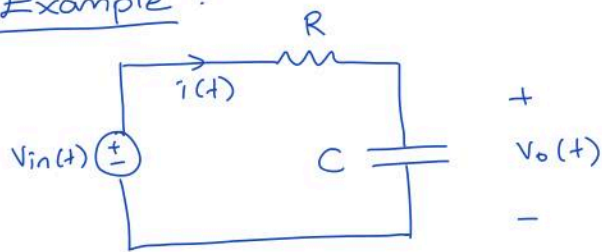
High Pass Filter (HPF)

* High frequencies "pass" almost unaffected.

* At low frequencies, the output reaches zero.

* Special case : when $\omega_0 = \frac{R}{L} \Rightarrow H(\omega) = \frac{j}{1+j}$

$$|H(\omega_0)| = \frac{1}{\sqrt{2}} = \frac{|H(\omega)|_{\max}}{\sqrt{2}}$$

Example :

$$V_o = V_{in} \cdot \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \quad (\text{voltage divider})$$

$$= V_{in} \cdot \frac{1}{1 + j\omega RC}$$

$$H(\omega) = \frac{V_o}{V_{in}} = \frac{1}{1 + j\omega RC}$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}} \Rightarrow \text{LPF}$$

$$* |H(\omega)|_{\max} = 1 \Rightarrow \text{at } \omega = 0$$

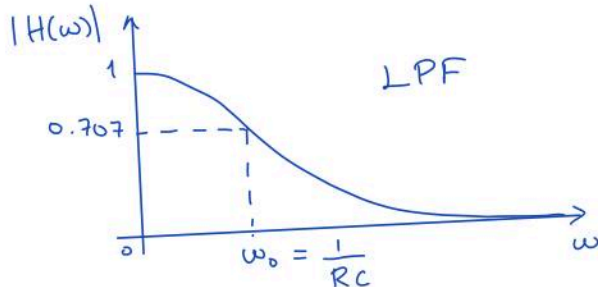
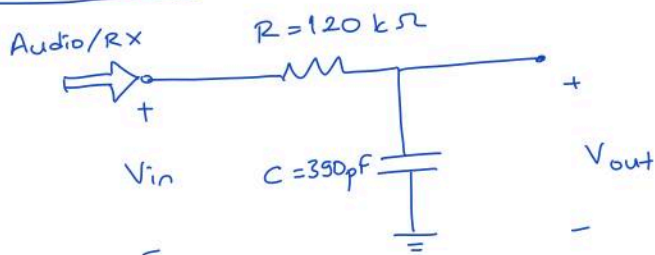
$$* \text{cut-off frequency: } |H(\omega_0)| = \frac{|H(\omega)|_{\max}}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + \omega_0^2 R^2 C^2}}$$

$$\omega_0^2 R^2 C^2 = 1$$

$$\omega_0 = \frac{1}{RC}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi RC}$$

Example :

3dB cut-off frequency:

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi RC}$$

$$= \frac{1}{2\pi \cdot 120 \cdot 10^3 \cdot 330 \cdot 10^{-12}}$$

$$\approx 3.4 \text{ kHz}$$

Limits the received audio signal to frequencies below $\sim 3.4 \text{ kHz}$

Logarithmic Plot :

$$H(\omega) = \frac{1}{1+j\omega RC} \quad \omega_0 = \frac{1}{RC}$$

$$H(\omega) = \frac{1}{1+j\frac{\omega}{\omega_0}}$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + \frac{\omega^2}{\omega_0^2}}}$$

* if $\omega \ll \omega_0$, $|H(\omega)| \approx \frac{1}{\sqrt{1}} = 1 \Rightarrow$ asymptotic behavior at low freqs.
 $20 \cdot \log_{10}(1) = 0 \text{ dB}$

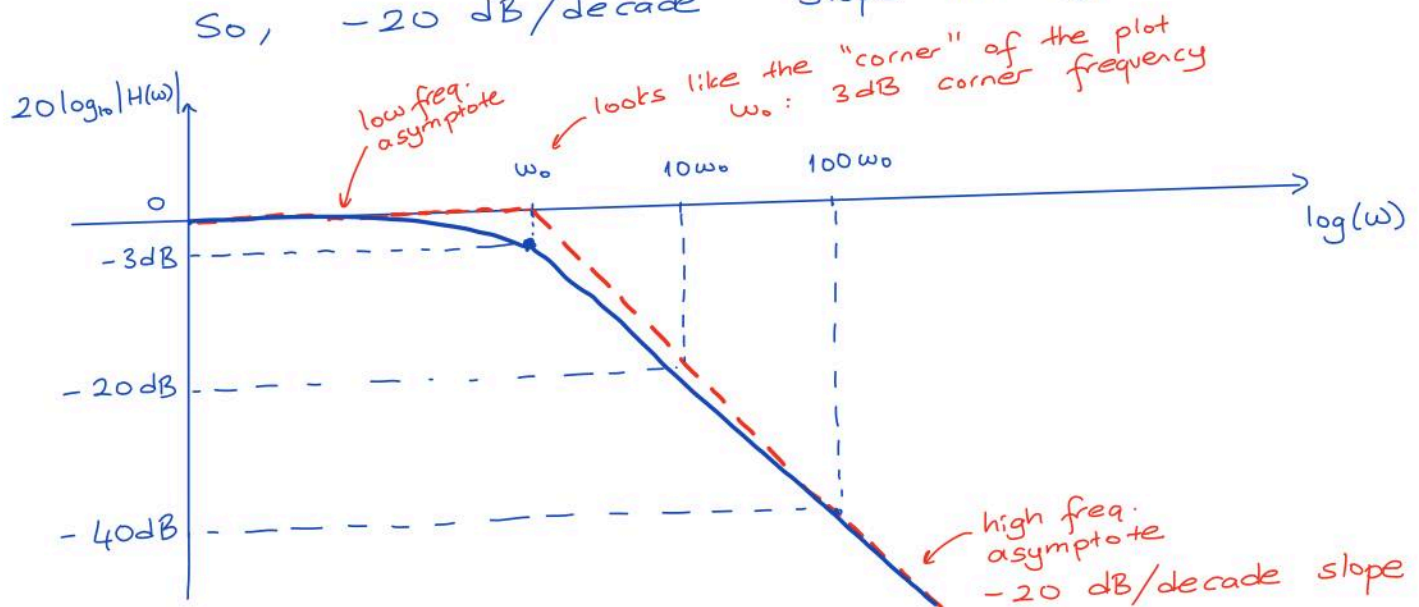
* at $\omega = \omega_0$, $|H(\omega_0)| = \frac{1}{\sqrt{2}} \Rightarrow 20 \log_{10}\left(\frac{1}{\sqrt{2}}\right) = -3 \text{ dB}$

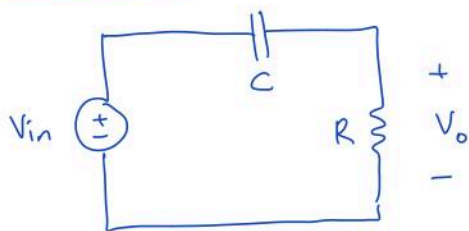
* if $\omega \gg \omega_0$, $|H(\omega)| \approx \frac{1}{\sqrt{\frac{\omega^2}{\omega_0^2}}} = \frac{\omega_0}{\omega} \Rightarrow$ asymptotic behavior at high freqs.

1 "decade" $\left\{ \begin{array}{l} @ \omega = \omega_0 \Rightarrow 20 \log_{10}\left(\frac{\omega_0}{\omega_0}\right) = 0 \text{ dB} \\ @ \omega = 10\omega_0 \Rightarrow 20 \log_{10}\left(\frac{\omega_0}{10\omega_0}\right) = 20 \log_{10}\left(\frac{1}{10}\right) = -20 \log_{10}(10) = -20 \text{ dB} \end{array} \right.$

1 "decade" $\left\{ \begin{array}{l} @ \omega = 100\omega_0 \Rightarrow 20 \log_{10}\left(\frac{\omega_0}{100\omega_0}\right) = -40 \text{ dB} \end{array} \right.$

So, -20 dB/decade slope in logarithmic plot.



Example :

$$H(\omega) = \frac{V_o}{V_{in}} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega RC}{1 + j\omega RC}$$

$$|H(\omega)| = \frac{\omega RC}{\sqrt{1 + \omega^2 R^2 C^2}}$$

$$\text{@ } \omega = 0 \Rightarrow |H(\omega)| = 0$$

$$\text{@ } \omega \rightarrow \infty \Rightarrow |H(\omega)| \rightarrow 1 = |H(\omega)|_{\max}$$

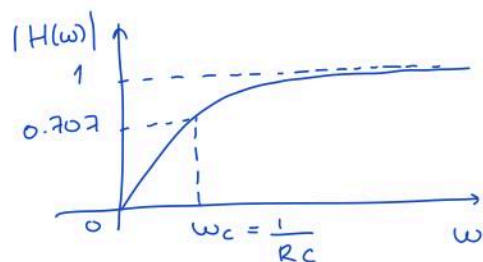
$$\text{* cut-off freq: } |H(\omega_c)| = \frac{|H(\omega)|_{\max}}{\sqrt{2}} = \frac{\omega_c RC}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

$$2\omega_c^2 R^2 C^2 = 1 + \omega_c^2 R^2 C^2$$

$$\omega_c RC = 1 \Rightarrow \omega_c = \frac{1}{RC}$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi RC}$$

$$\text{@ } \omega = \omega_c = \frac{1}{RC} : H(\omega_c) = \frac{j}{1+j}$$

Logarithmic Plot

$$H(\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

$$\omega_c = \frac{1}{RC}$$

$$H(\omega) = \frac{j \frac{\omega}{\omega_c}}{1 + j \frac{\omega}{\omega_c}}$$

$$|H(\omega)| = \frac{\frac{\omega}{\omega_c}}{\sqrt{1 + \frac{\omega^2}{\omega_c^2}}}$$

$$@ \omega = \omega_c \Rightarrow |H(\omega_c)| = \frac{1}{\sqrt{2}} \Rightarrow -3\text{dB}$$

$$* \text{ If } \omega \ll \omega_c \Rightarrow |H(\omega)| \approx \frac{\omega}{\omega_c} \Rightarrow \text{low frequency asymptote}$$

one decade $\nearrow$ $@ \omega = \omega_c, |H(\omega_c)| = 1 \Rightarrow 0\text{dB}$ $\nwarrow +20\text{dB}$

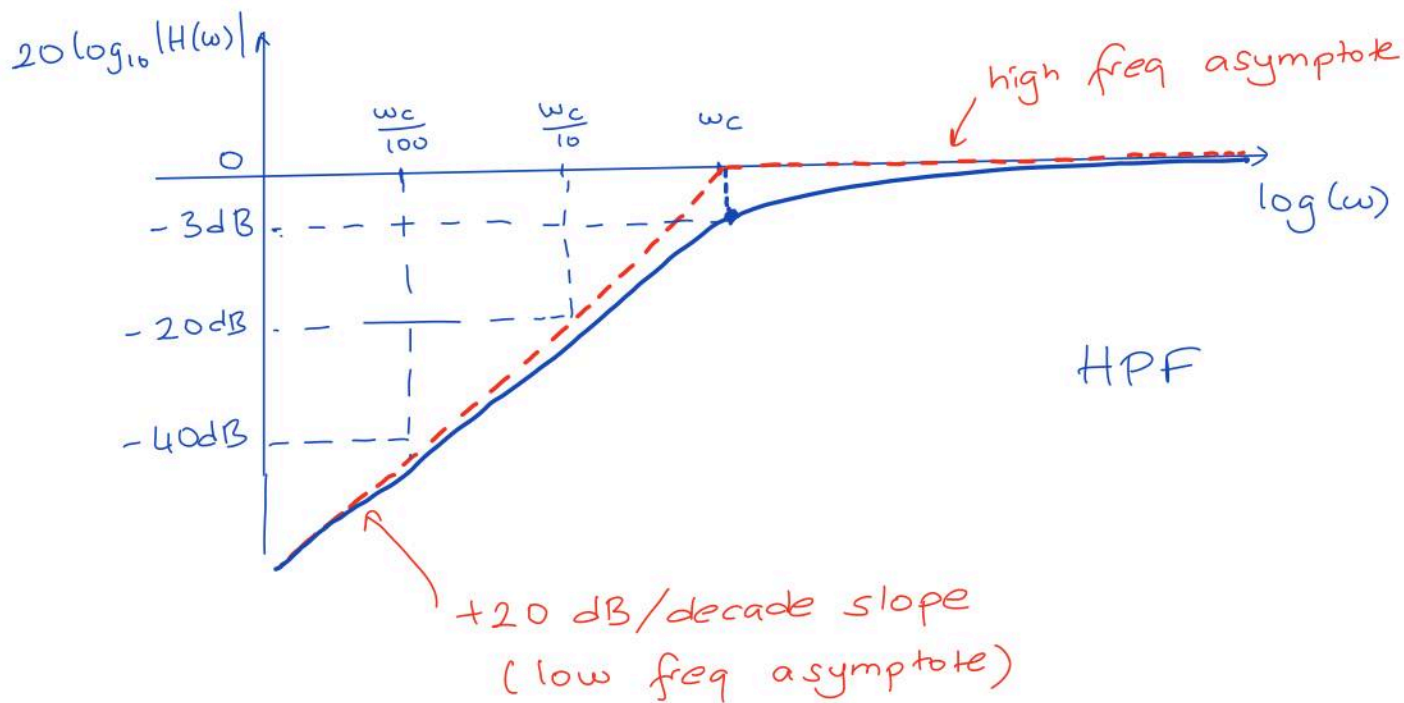
one decade $\nearrow$ $@ \omega = \frac{\omega_c}{10}, |H(\frac{\omega_c}{10})| = \frac{\frac{\omega_c}{10}}{\omega_c} = \frac{1}{10} \Rightarrow -20\text{dB}$ $\nwarrow +20\text{dB}$

$@ \omega = \frac{\omega_c}{100}, |H(\frac{\omega_c}{100})| = \frac{1}{100} \Rightarrow -40\text{dB}$ $\nwarrow +20\text{dB}$

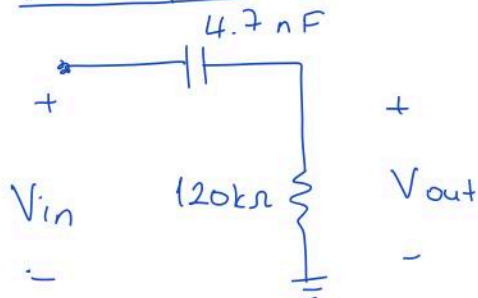
So, $+20\text{ dB/decade}$ slope

$$* \text{ If } \omega \gg \omega_c \Rightarrow |H(\omega)| \approx \frac{\frac{\omega}{\omega_c}}{\sqrt{\frac{\omega^2}{\omega_c^2}}} = 1 \Rightarrow \text{high frequency asymptote}$$

$$\Rightarrow 0\text{dB}$$



Example:



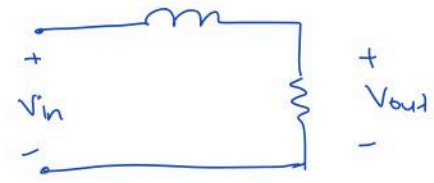
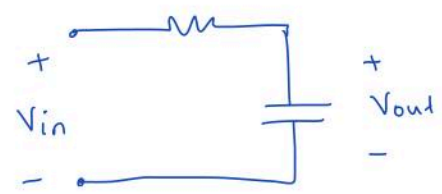
3-dB cut-off freq:

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi RC}$$

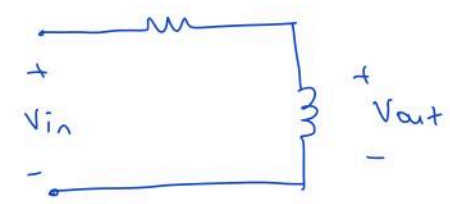
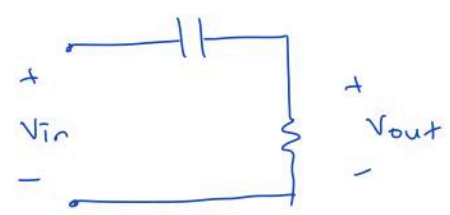
$$= \frac{1}{2\pi \times 120 \times 10^3 \times 4.7 \times 10^{-9}}$$

$$\approx 282\text{ Hz}$$

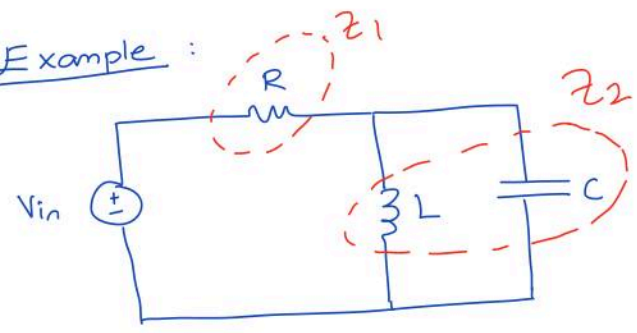
LPF :



HPF :



Example :



+
V_{out}
-

$$H(\omega) = \frac{V_{out}}{V_{in}} = ?$$

$$V_{out} = V_{in} \cdot \frac{Z_2}{Z_1 + Z_2}$$

$$\Rightarrow H(\omega) = \frac{V_{out}}{V_{in}} = \frac{Z_2}{Z_1 + Z_2}$$

$$H(\omega) = \frac{(j\omega L \parallel \frac{1}{j\omega C})}{R + (j\omega L \parallel \frac{1}{j\omega C})} = \frac{(\frac{1}{j\omega L} + j\omega C)^{-1}}{R + (\frac{1}{j\omega L} + j\omega C)^{-1}} = \frac{\frac{j\omega L}{1 - \omega^2 LC}}{R + \frac{j\omega L}{1 - \omega^2 LC}}$$

$$= \frac{j\omega L}{R(1 - \omega^2 LC) + j\omega L}$$

* when $\omega^2 LC = 1 \Rightarrow \omega = \frac{1}{\sqrt{LC}} \Rightarrow H(\omega) = 1 \Rightarrow V_{out} = V_{in}$

Why use log scale?

We perceive sound in logarithmic scale.

Total silence $\rightarrow$ 0 dB

Sound of breathing $\rightarrow$ 10 dB

Normal conversation $\rightarrow$ 50 dB

Vacuum cleaner $\rightarrow$ 70 dB

Rock concert $\rightarrow$ 120 dB

Threshold of pain $\rightarrow$ 130 dB

Nearby jet plane $\rightarrow$ 150 dB

Death of hearing tissue $\rightarrow$ 180 dB

Loudest sound possible $\rightarrow$ 194 dB

10^7 times
higher
intensity!!

Pressure amplitude
is 20 dB larger
 $\Rightarrow$ 10 times larger

Intensity $\propto A^2$
so, 100 times higher
intensity!!

$\Rightarrow$ we perceive it as
 ~ 4 times louder

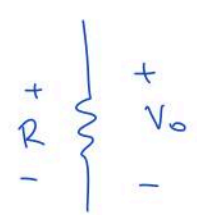
- * 140 dB or higher causes irreversible nerve damage
- > 85 dB $\rightarrow$ reduces hearing acuity over time

Decibel for voltage gain vs. Power gain :

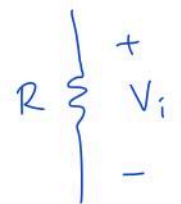
Voltage gain : $A = \frac{V_o}{V_i} \Rightarrow A_{dB} = 20 \log_{10} \left(\frac{V_o}{V_i} \right)$

Power gain : $G = \frac{P_o}{P_i} \Rightarrow G_{dB} = 10 \log_{10} \left(\frac{P_o}{P_i} \right)$

*



$$P_o = \frac{V_o^2}{R}$$



$$P_i = \frac{V_i^2}{R}$$

$$A = \frac{V_o}{V_i}$$

$$G = \frac{P_o}{P_i} = \frac{\frac{V_o^2}{R}}{\frac{V_i^2}{R}} = \frac{V_o^2}{V_i^2}$$

$$G_{dB} = 10 \log_{10} \left(\frac{V_o^2}{V_i^2} \right) = 20 \log_{10} \left(\frac{V_o}{V_i} \right) = A_{dB}$$

* So, we get the same value in dB both ways.

$$-3dB \Rightarrow \frac{V_o}{V_i} = \frac{1}{\sqrt{2}} \quad \text{OR} \quad \frac{P_o}{P_i} = \frac{1}{2}$$

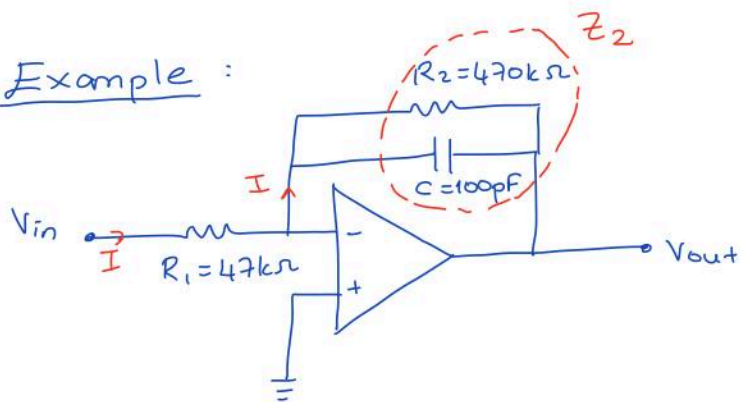
* Decibel is also used to define absolute levels.

1 mW $\Rightarrow$ 0 dBm "m" denotes that this value is with respect to 1 milliWatt power

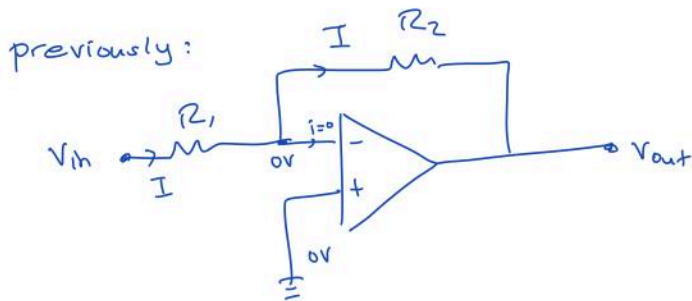
$$0.5 \text{ mW} \Rightarrow 10 \log_{10} \left(\frac{0.5 \text{ mW}}{1 \text{ mW}} \right) = 10 \log_{10} \left(\frac{1}{2} \right) = -3 \text{ dBm}$$

$$\begin{aligned} 20 \text{ W} &\Rightarrow 10 \log_{10} \left(\frac{20 \text{ W}}{1 \text{ mW}} \right) = 10 \log_{10} \left(\frac{20 \cdot 10^3}{1} \right) \\ &= 10 \log_{10}(2) + 10 \log_{10}(10^4) \\ &= 3 + 40 = 43 \text{ dBm} \end{aligned}$$

Example :



Assume V_{in} is sinusoidal.
Find and plot $H(\omega)$,
Assume OPAMP is not saturated.



$$V_{out} = -\frac{R_2}{R_1} \cdot V_{in}$$

$$I = \frac{V_{in} - 0}{R_1} = \frac{0 - V_{out}}{R_2}$$

Now : $V_{out} = -\frac{Z_2}{R_1} \cdot V_{in}$ (from inverting amplifier)

$$H(\omega) = \frac{V_{out}}{V_{in}} = -\frac{Z_2}{R_1} = -\frac{R_2 // C}{R_1} = -\frac{\left(\frac{1}{R_2} + j\omega C\right)^{-1}}{R_1}$$

$$= -\frac{\frac{R_2}{1 + j\omega R_2 C}}{R_1} = \underbrace{-\frac{R_2}{R_1}}_{\text{gain}} \cdot \underbrace{\frac{1}{1 + j\omega R_2 C}}_{\text{low pass filter}}$$

$$|H(\omega)| = \frac{R_2}{R_1} \cdot \frac{1}{\sqrt{1 + \omega^2 R_2^2 C^2}}$$

@ $\omega = 0$, $|H(\omega)|_{\max} = H(0) = \frac{R_2}{R_1} = 10$

@ $\omega \rightarrow \infty$, $|H(\omega)| \rightarrow 0$

@ cut-off freq: $|H(\omega_c)| = \frac{|H(\omega)|_{\max}}{\sqrt{2}} = \frac{R_2}{R_1} \cdot \frac{1}{\sqrt{2}} = \frac{10}{\sqrt{2}}$

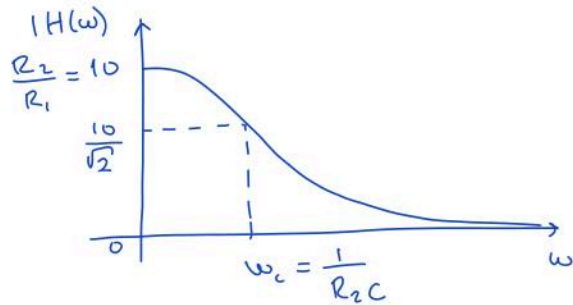
$$\frac{R_2}{R_1} \cdot \frac{1}{\sqrt{1 + \omega_c^2 R_2^2 C^2}} = \frac{R_2}{R_1} \cdot \frac{1}{\sqrt{2}}$$

$$\omega_c = \frac{1}{R_2 C}$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi R_2 C} = \frac{1}{2\pi \times 470 \times 10^3 \times 100 \times 10^{-12}} = 3.38 \text{ kHz}$$

$$* \quad H(\omega) = -\frac{R_2}{R_1} \cdot \frac{1}{1 + j\omega R_2 C} = -\frac{R_2}{R_1} \cdot \frac{1}{1 + j\frac{\omega}{\omega_c}}$$

$$|H(\omega)| = \frac{R_2}{R_1} \cdot \frac{1}{\sqrt{1 + \frac{\omega^2}{\omega_c^2}}}$$

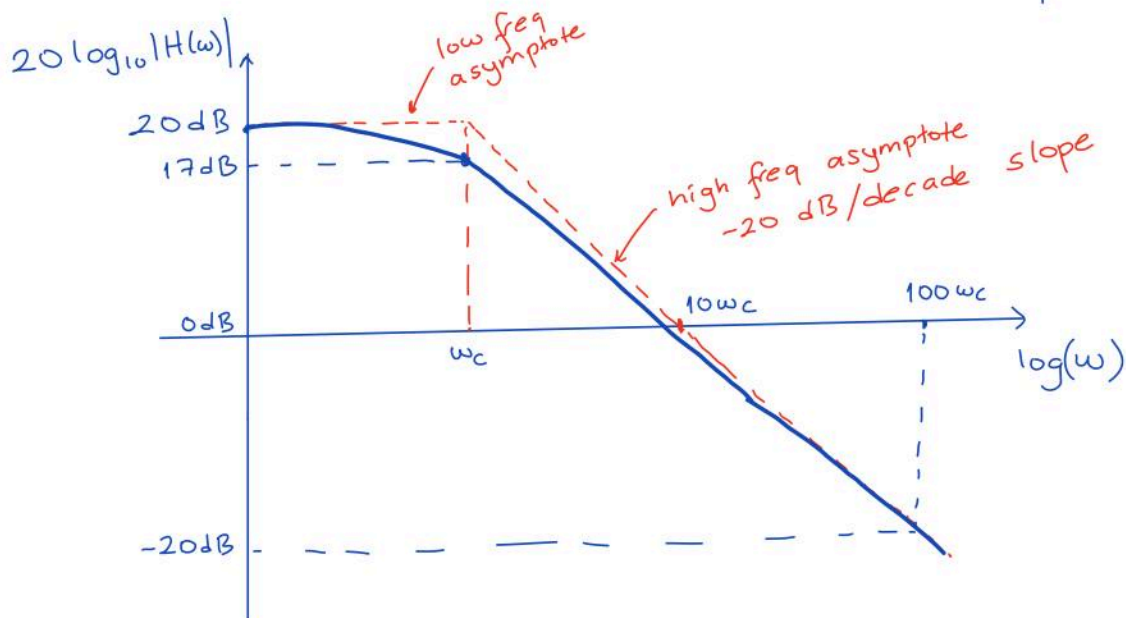
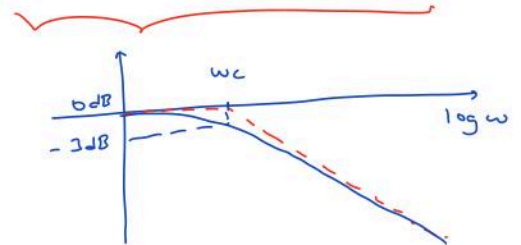


* In logarithmic scale

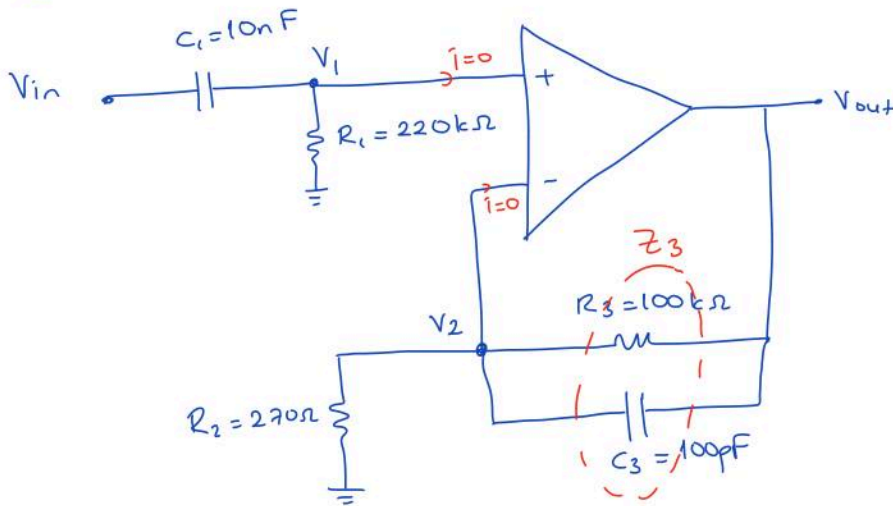
$$20 \log_{10} |H(\omega)| = 20 \log_{10} \left(\frac{R_2}{R_1} \cdot \frac{1}{\sqrt{1 + \frac{\omega^2}{\omega_c^2}}} \right)$$

$$= 20 \log_{10} \left(\frac{R_2}{R_1} \right) + 20 \log_{10} \left(\frac{1}{\sqrt{1 + \frac{\omega^2}{\omega_c^2}}} \right)$$

$\underbrace{\hspace{10em}}_{20 \text{ dB}}$



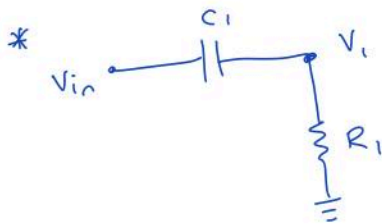
Example: Band pass filter



Assume OPAMP is not saturated.

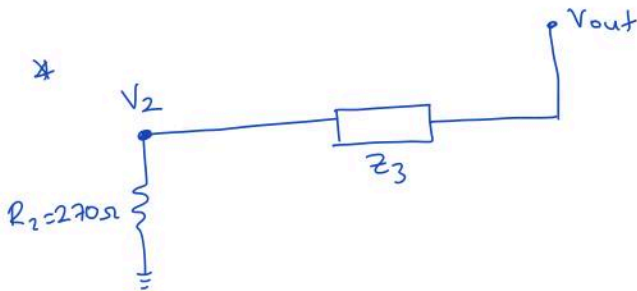
$$V_1 = V_2$$

$$\text{Find } H(\omega) = \frac{V_{out}}{V_{in}}$$



$$V_1 = V_{in} \cdot \frac{R_1}{R_1 + \frac{1}{j\omega C_1}} = V_{in} \cdot \frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1}$$

(voltage divider)



$$V_2 = V_{out} \cdot \frac{R_2}{R_2 + Z_3}$$

* Finally, $V_1 = V_2$

$$V_{in} \cdot \frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1} = V_{out} \cdot \frac{R_2}{R_2 + Z_3}$$

$$H(\omega) = \frac{V_{out}}{V_{in}} = \left(\frac{R_2 + Z_3}{R_2} \right) \cdot \frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1}$$

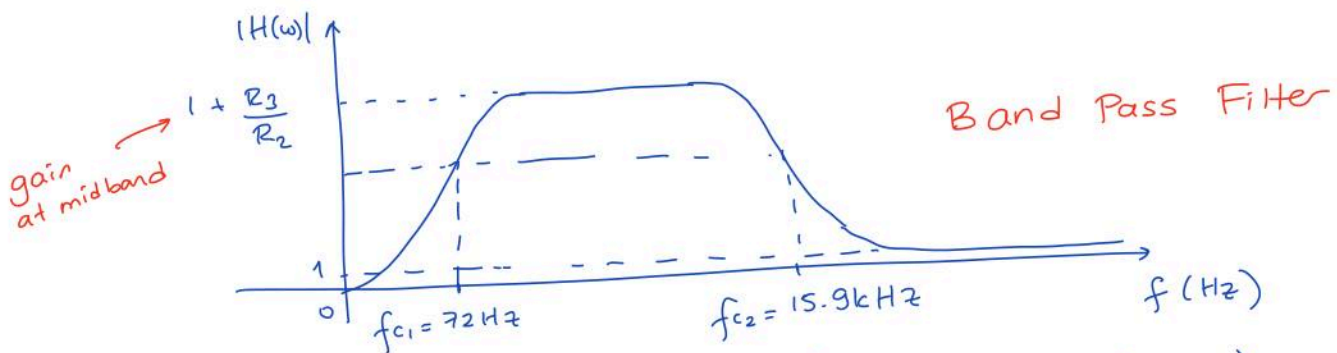
$$= \left(1 + \frac{Z_3}{R_2} \right) \cdot \frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1}$$

$$Z_3 = R_3 \parallel \frac{1}{j\omega C_3} = \left(\frac{1}{R_3} + j\omega C_3 \right)^{-1} = \frac{R_3}{1 + j\omega R_3 C_3}$$

$$H(\omega) = \left(1 + \underbrace{\frac{R_3}{R_2}}_{\approx 370} \cdot \underbrace{\frac{1}{1 + j\omega R_3 C_3}}_{\text{LPF}} \right) \cdot \underbrace{\left(\frac{-j\omega R_1 C_1}{1 + j\omega R_1 C_1} \right)}_{\text{HPF}}$$

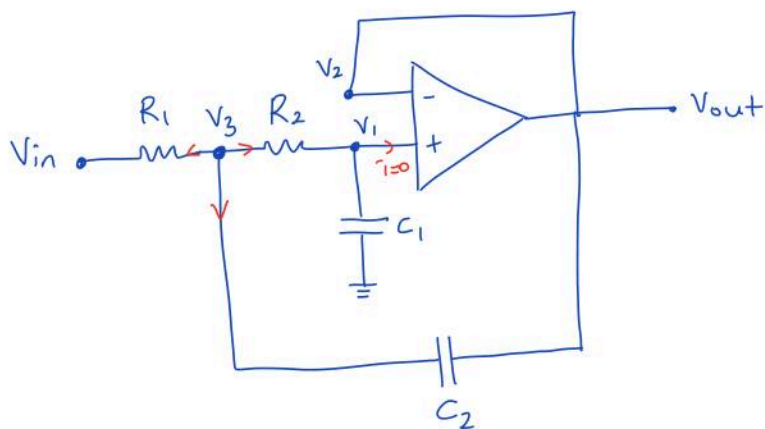
$$f_{c2} = \frac{\omega_c}{2\pi} = \frac{1}{2\pi R_3 C_3} = 15.9 \text{ kHz}$$

$$f_{c1} = \frac{\omega_c}{2\pi} = \frac{1}{2\pi R_1 C_1} = 72 \text{ Hz}$$



$$\text{Gain at midband: } 20 \log_{10} \left(1 + \frac{R_3}{R_2} \right) = 20 \log_{10} (371) \approx 51.4 \text{ dB}$$

Example: 2nd order LPF



Assume OPAMP is not saturated $\Rightarrow V_1 = V_2$

$$V_2 = V_{out}$$

$$V_1 = V_2 = V_{out}$$

KCL at V_3 : $\frac{V_3 - V_{in}}{R_1} + \frac{V_3 - V_{out}}{\frac{1}{j\omega C_2}} + \frac{V_3 - V_{out}}{R_2} = 0$

Voltage divider at V_1 : $V_1 = V_3 \cdot \frac{\frac{1}{j\omega C_1}}{R_2 + \frac{1}{j\omega C_1}} = V_{out}$

$$V_{out} = V_3 \cdot \frac{1}{1 + j\omega R_2 C_1}$$

$$V_3 = V_{out} (1 + j\omega R_2 C_1)$$

Insert into the first equation:

$$H(\omega) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + j\omega C_1 (R_1 + R_2) - \omega^2 R_1 R_2 C_1 C_2}$$

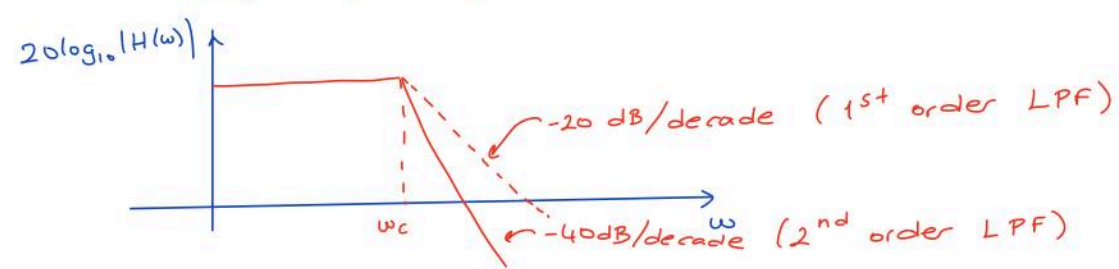
* at VERY high frequencies:

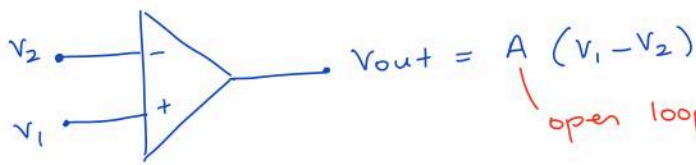
$$|H(\omega)| \approx \frac{1}{\omega^2 R_1 R_2 C_1 C_2}$$

$$\begin{matrix} \omega_1 & \longrightarrow & 10\omega_1 \\ |H(\omega_1)| & \longrightarrow & \frac{|H(\omega_1)|}{100} \end{matrix}$$

(in one decade)
(goes down by 40 dB)

$\Rightarrow$ -40 dB/decade slope at high frequencies

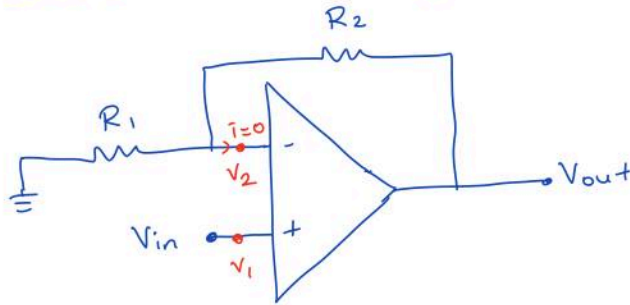


OPAMP :

open loop voltage gain. $A \rightarrow \infty$ (for ideal OPAMP)

If v_{out} is finite and OPAMP is not saturated,
 $v_1 = v_2$

* where did "A" go in the examples we have seen?



$$v_{out} = A(v_1 - v_2)$$

$$v_1 = v_{in}$$

$$v_2 = v_{out} \cdot \frac{R_1}{R_1 + R_2} \quad (\text{voltage divider})$$

So,

$$v_{out} = A(v_1 - v_2)$$

$$= A\left(v_{in} - v_{out} \cdot \frac{R_1}{R_1 + R_2}\right)$$

$$v_{out} \left(1 + \frac{A \cdot R_1}{R_1 + R_2}\right) = A \cdot v_{in}$$

$$v_{out} = v_{in} \cdot \frac{A(R_1 + R_2)}{(R_1 + R_2) + A \cdot R_1}$$

* If $A \rightarrow \infty \Rightarrow v_{out} = v_{in} \cdot \frac{(R_1 + R_2)}{R_1} = \underbrace{\left(1 + \frac{R_2}{R_1}\right)}_{\text{same as before}} \cdot v_{in}$

Now, check: $v_2 = v_{out} \cdot \frac{R_1}{R_1 + R_2} = \left(1 + \frac{R_2}{R_1}\right) v_{in} \cdot \frac{R_1}{R_1 + R_2}$

$$v_2 = v_{in} \Rightarrow \boxed{v_1 = v_2}$$

* So, the feedback loop forces output to be finite such that $v_1 = v_2$ (when OPAMP is not saturated).

* We can get the same result by starting with $v_1 = v_2$.

Calculating Power Using Phasors

(34)

* In general : $p(t) = v(t) \cdot i(t)$

* For DC : $P = V \cdot I$

* For AC (sinusoidal) : $P = \frac{V \cdot I^*}{2}$ (V and I are in phasor notation)
complex valued

Dissipated power : $P_d = \operatorname{Re}\{P\} = \operatorname{Re}\left\{\frac{V \cdot I^*}{2}\right\}$

$$V = V_p \cdot e^{j\phi_1}$$

$$I = I_p \cdot e^{j\phi_2}$$

$$P_d = \operatorname{Re}\left\{\frac{V \cdot I^*}{2}\right\} = \operatorname{Re}\left\{\frac{V}{\sqrt{2}} \cdot \frac{I^*}{\sqrt{2}}\right\}$$

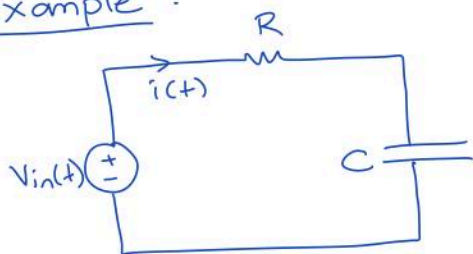
$$= \operatorname{Re}\left\{\frac{V_p \cdot e^{j\phi_1}}{\sqrt{2}} \cdot \frac{I_p e^{-j\phi_2}}{\sqrt{2}}\right\} = \operatorname{Re}\left\{\underbrace{\frac{V_p}{\sqrt{2}}}_{V_{rms}} \cdot \underbrace{\frac{I_p}{\sqrt{2}}}_{I_{rms}} e^{j(\phi_1 - \phi_2)}\right\}$$

phase diff. between voltage and current

$$= \frac{V_p}{\sqrt{2}} \cdot \frac{I_p}{\sqrt{2}} \cos(\phi_1 - \phi_2)$$

* $\operatorname{Im}\left\{\frac{V \cdot I^*}{2}\right\}$ is called reactive power

Example :



$V_{in}(t)$ is sinusoidal.

Compute total power dissipated in this circuit.

$$I = \frac{V_{in}}{R + \frac{1}{j\omega C}} = V_{in} \cdot \frac{j\omega C}{1 + j\omega RC}$$

$$P_d = \operatorname{Re}\left\{\frac{V_{in} \cdot I^*}{2}\right\} = \operatorname{Re}\left\{\frac{V_{in} \cdot V_{in}^*}{2} \left(\frac{-j\omega C}{1 - j\omega RC}\right)\right\}$$
$$= \operatorname{Re}\left\{\frac{|V_{in}|^2}{2} \cdot \frac{-j\omega C}{1 - j\omega RC} \right\} (1 + j\omega RC)$$

$$= \operatorname{Re} \left\{ \frac{|V_{in}|^2}{2} \cdot \frac{\omega^2 R C^2 - j\omega C}{1 + \omega^2 R^2 C^2} \right\}$$

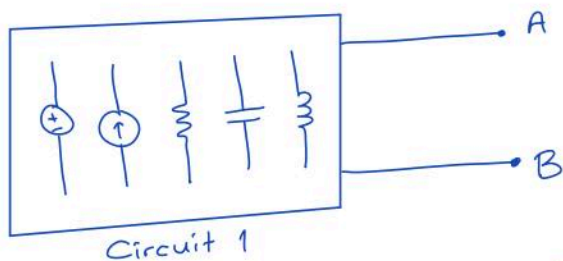
$$= \frac{|V_{in}|^2}{2} \cdot \frac{\omega^2 R C^2}{1 + \omega^2 R^2 C^2}$$

Question: What is P_d for resistor only? $\Rightarrow$ should be the same

What is P_d for capacitor only? $\Rightarrow$ should be zero

Equivalent Circuits:

Let's say we have a circuit made up of linear elements:



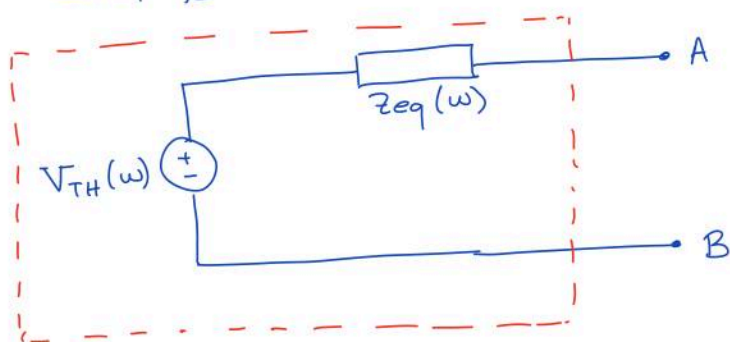
Linear circuits can be simplified for the sake of circuit analysis

Two different ways to simplify:

- 1) Thevenin equivalent circuit
- 2) Norton equivalent circuit

Thevenin Equivalent Circuit

Simplify the circuit to:



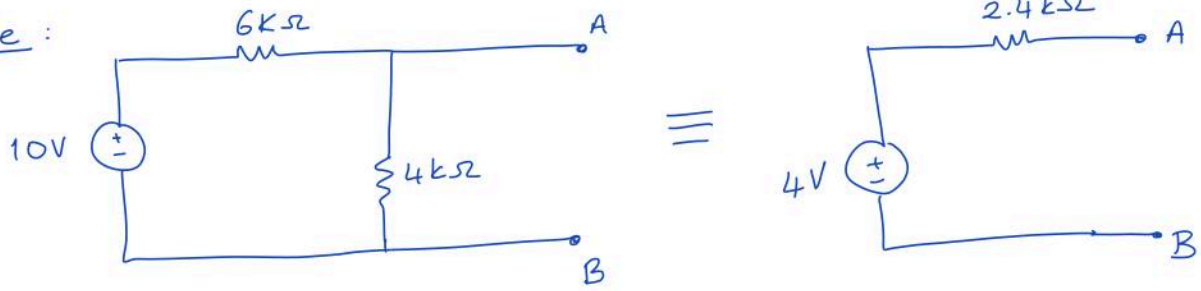
To find V_{TH} :

V_{TH} = open-circuit voltage across A-B

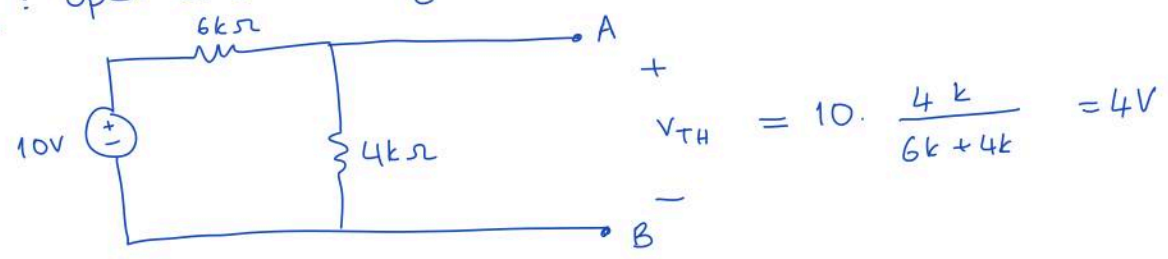
To find Z_{eq} :

Kill all sources. Find the impedance between A-B.

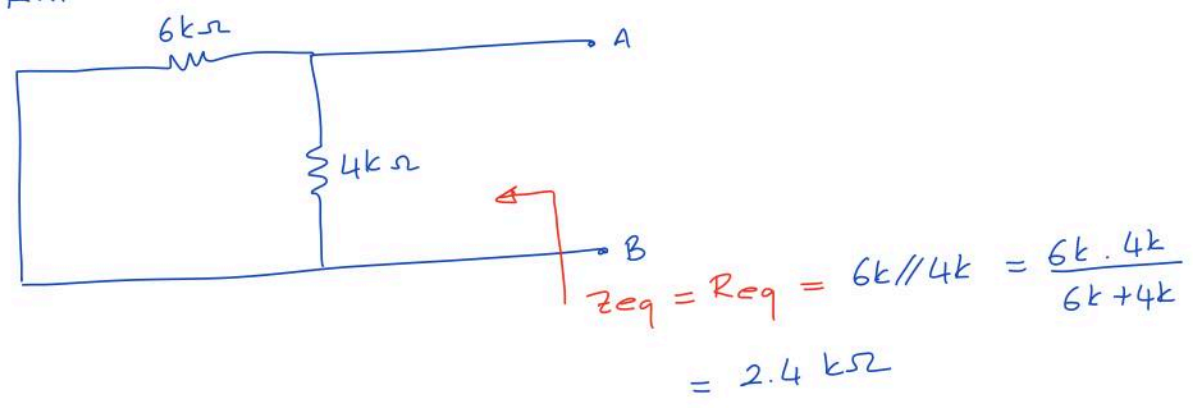
Example :



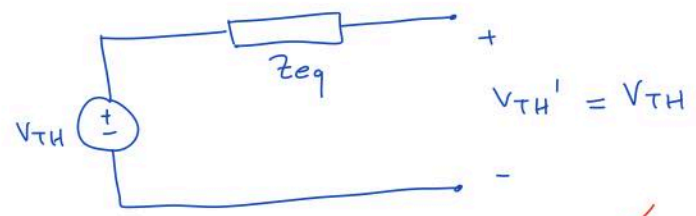
V_{TH} : open-circuit voltage across A-B



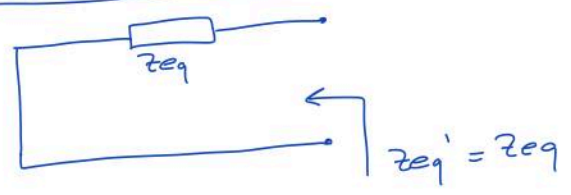
Z_{eq} : kill all sources.



* Thevenin equivalent of a Thevenin equivalent circuit?



kill all sources :

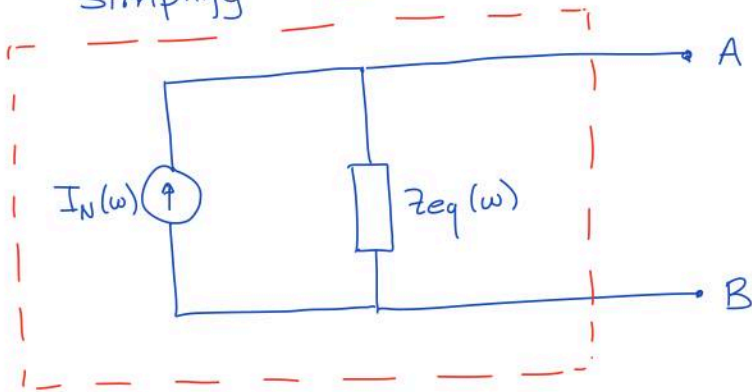


✓ consistent

Norton Equivalent Circuit :

(37)

Simplify the circuit to :

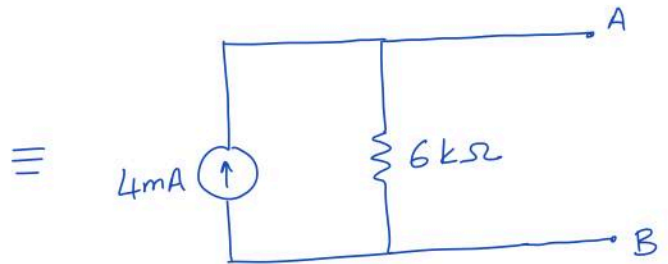
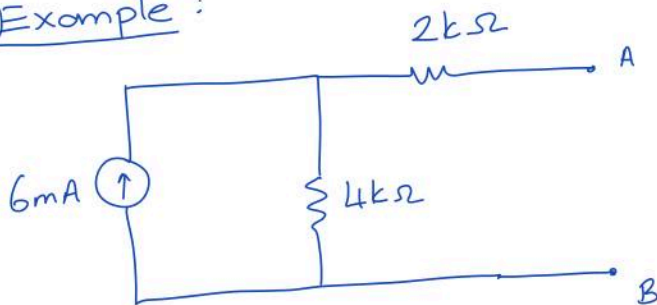


To find I_N :

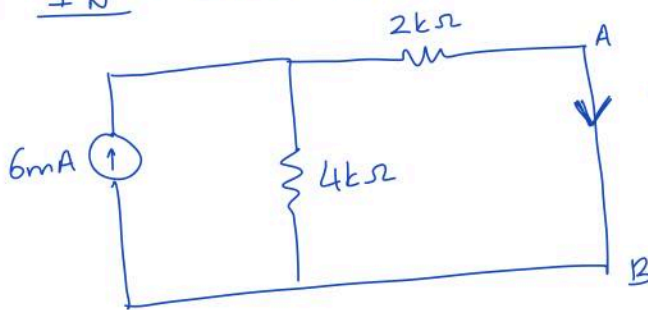
I_N = short-circuit current between A-B

To find Z_{eq} : (same as before)
kill all sources. Find the impedance between A-B.

Example :



* I_N : short-circuit current between A-B



$$I_N = 6\text{mA} \cdot \frac{4\text{k}}{4\text{k} + 2\text{k}} = 4\text{mA}$$

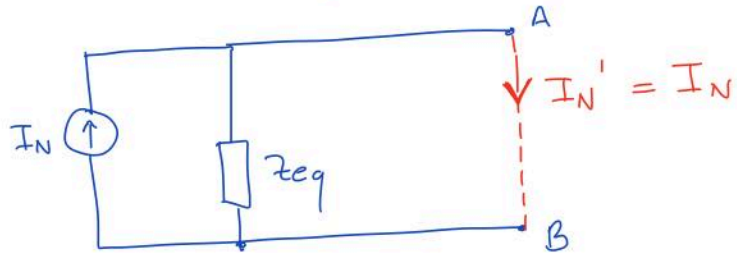
(current divider)

* Z_{eq} : kill all sources

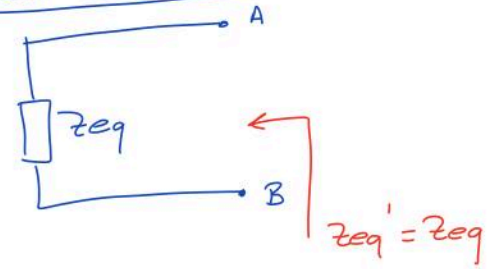


$$Z_{eq} = R_{eq} = 4\text{k} + 2\text{k} = 6\text{k}\Omega$$

* Norton equivalent of a Norton equivalent circuit?

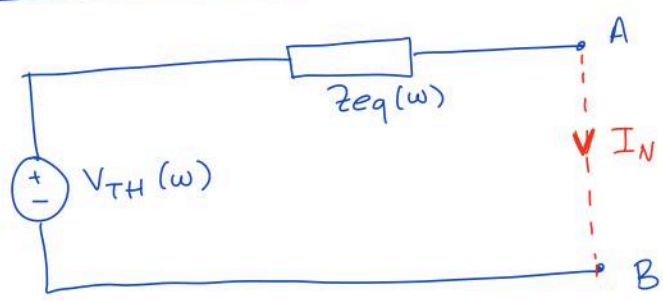


kill all sources

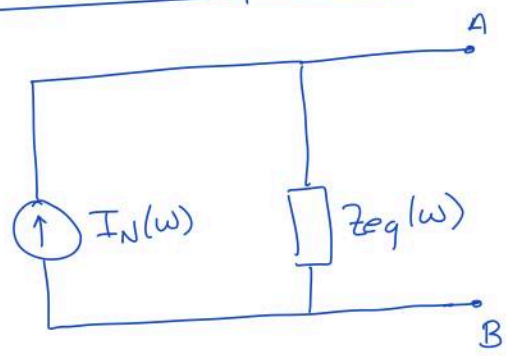


✓ consistent

* What is the Norton equivalent of a Thevenin equivalent circuit?



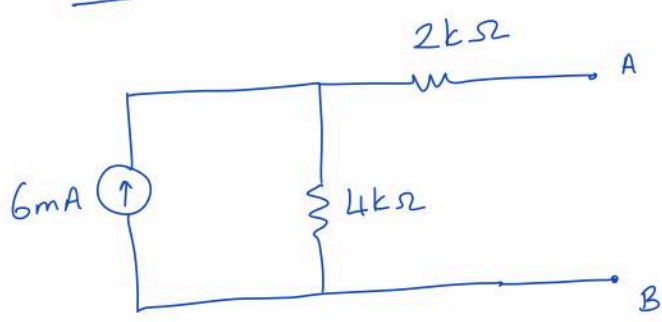
=



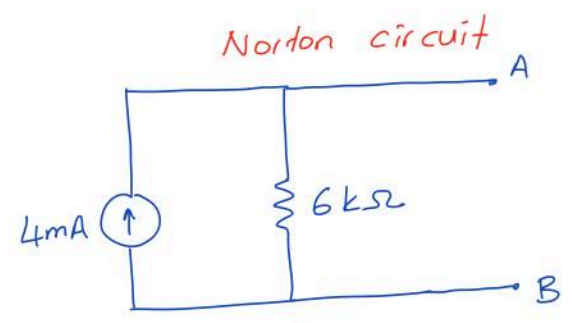
$$I_N = \frac{V_{TH}}{z_{eq}}$$

* So, if we know one, we can find the other easily.

Example: previous circuit

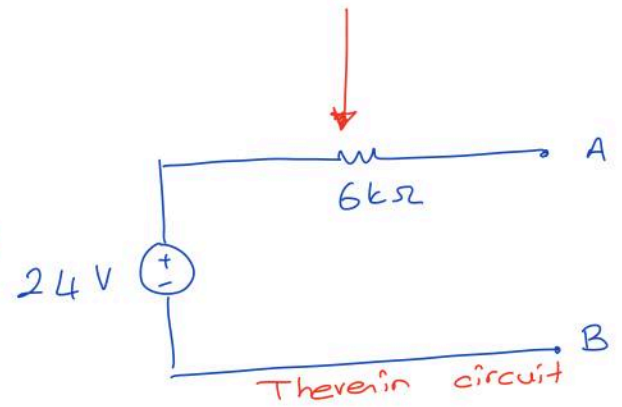


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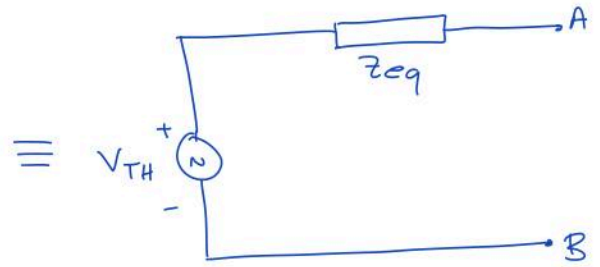
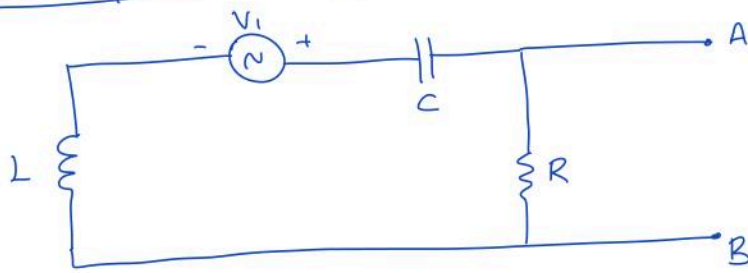
Norton circuit

exercise



Thevenin circuit

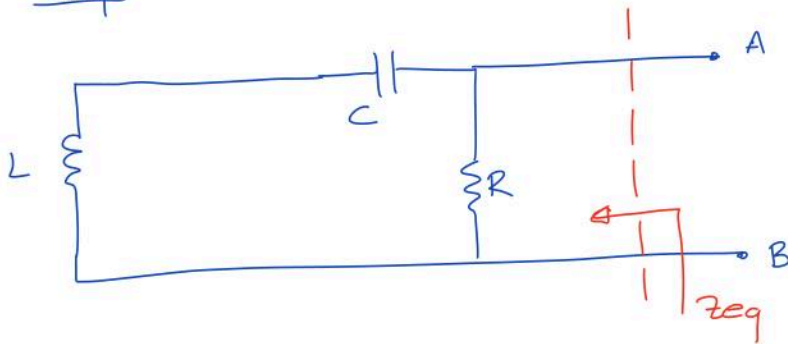
Example : Application to phasors



* V_{TH} : $V_{TH}(\omega) = V_i \cdot \frac{R}{R + \frac{1}{j\omega C} + j\omega L}$

$$= V_i \cdot \frac{j\omega R C}{1 - \omega^2 L C + j\omega R C}$$

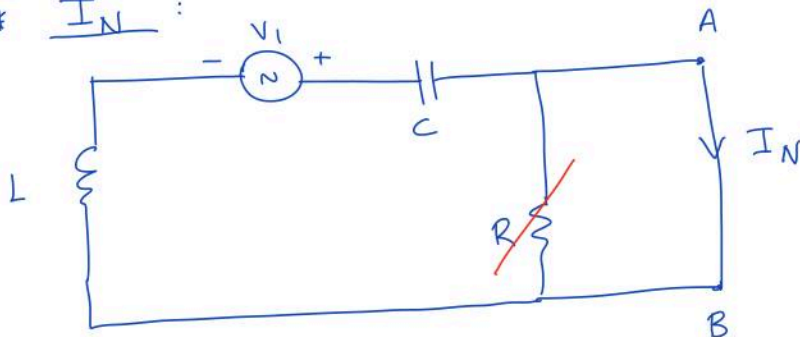
* Z_{eq} : kill all sources.



$$Z_{eq}(\omega) = R \parallel (j\omega L + \frac{1}{j\omega C}) = \frac{R \cdot (j\omega L + \frac{1}{j\omega C})}{R + j\omega L + \frac{1}{j\omega C}}$$

$$= \frac{R \cdot (1 - \omega^2 L C)}{1 - \omega^2 L C + j\omega R C}$$

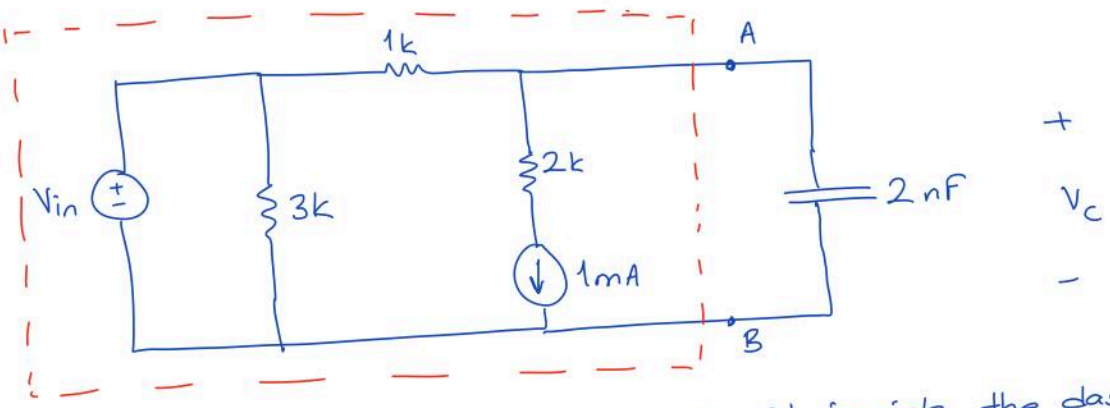
* I_N :



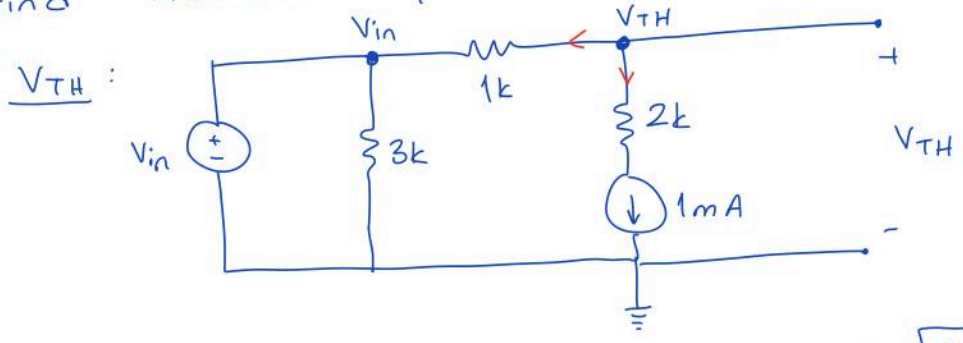
$$I_N(\omega) = \frac{V_i}{j\omega L + \frac{1}{j\omega C}} = V_i \cdot \frac{j\omega C}{1 - \omega^2 L C}$$

Check : $I_N(\omega) = \frac{V_{TH}(\omega)}{Z_{eq}(\omega)}$

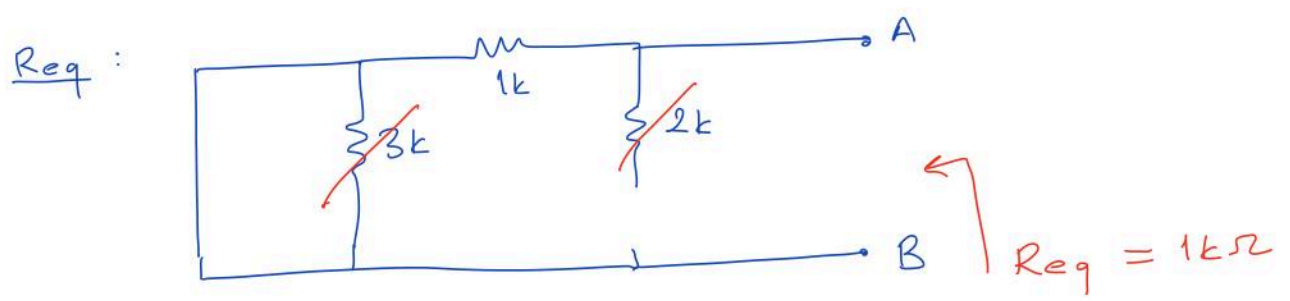
Example :



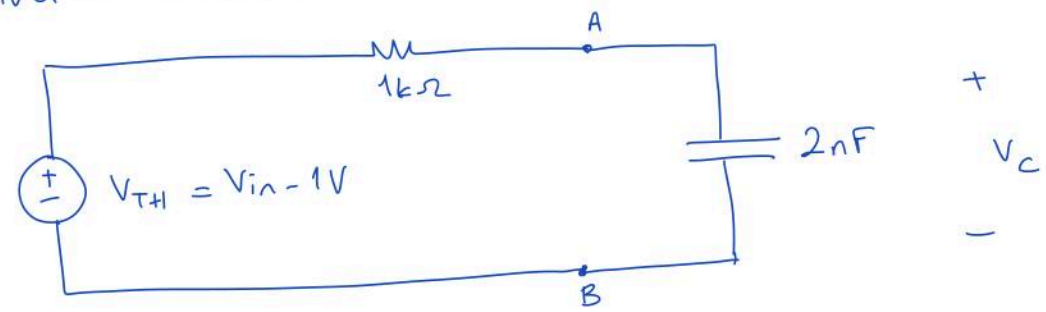
a) Find Thevenin equivalent of the circuit inside the dashed box.



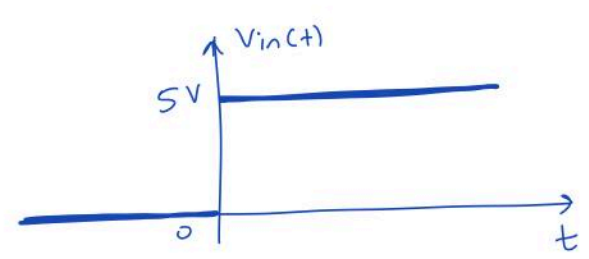
KCL : $\frac{V_{TH} - V_{in}}{1k\Omega} + 1mA = 0 \Rightarrow \boxed{V_{TH} = V_{in} - 1V}$



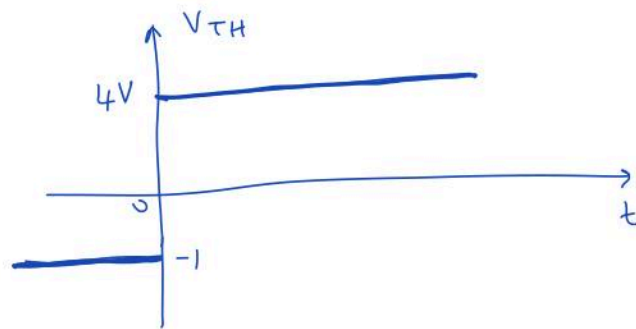
* Equivalent circuit :



b) Find $V_C(t)$ if $V_{in}(t)$ is as shown on the right.



Then,



$$V_{TH} = V_{in} - 1V$$

* for $t < 0$, $V_c(t) = -1V$

* $V_c(0^+) = V_c(0^-) = -1V = V_{init}$

* As $t \rightarrow \infty$, $V_{final} = 4V$

* So, for $t \geq 0$

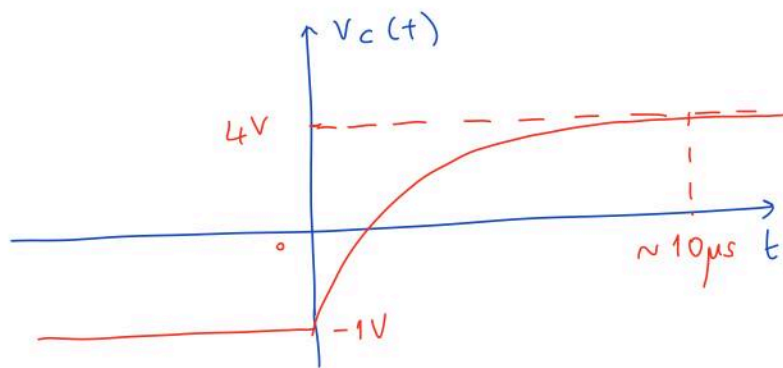
$$V_c(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau}$$

$$= 4 - 5e^{-t/\tau} V,$$

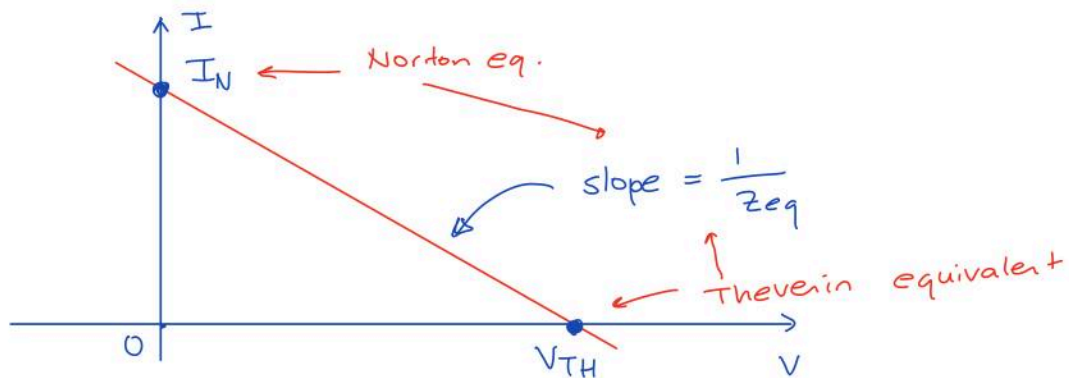
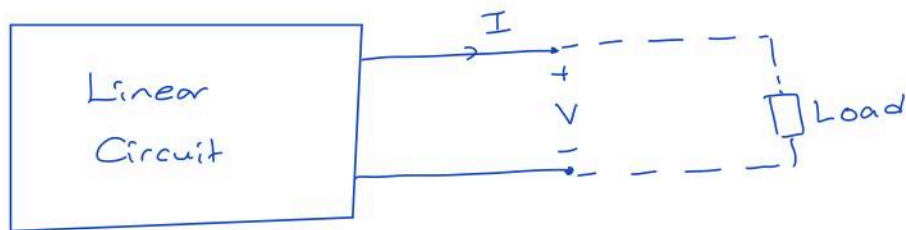
$$\tau = RC$$

$$\tau = 1k\Omega \times 2nF$$

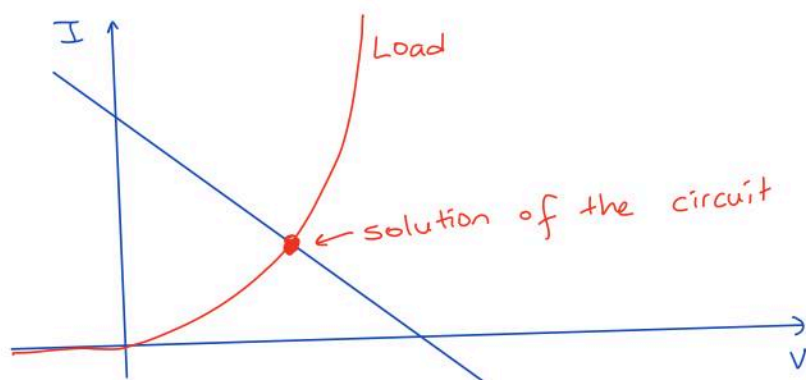
$$= 2\mu s$$



About Thevenin-Norton Equivalent Circuits



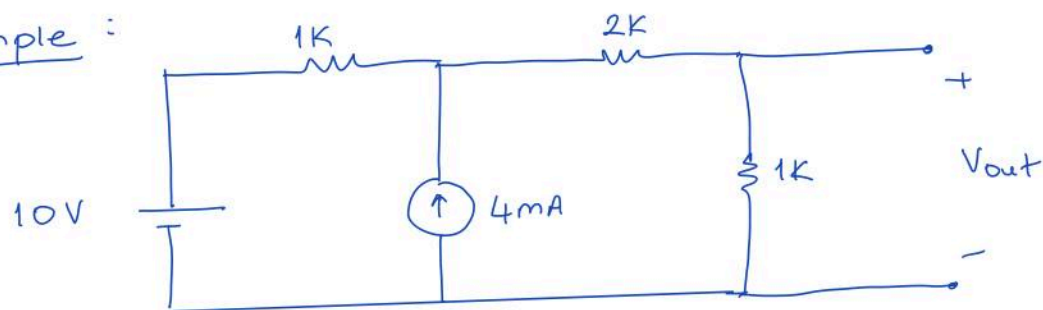
* Load may be non-linear



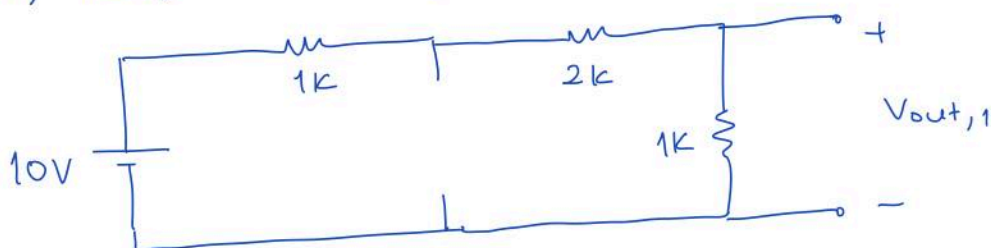
Superposition Principle : for linear circuits

- If a linear circuit has more than one independent source
- Kill all sources except one, then solve the circuit
- Repeat for all sources, one by one.
- Finally, add the results.

Example :

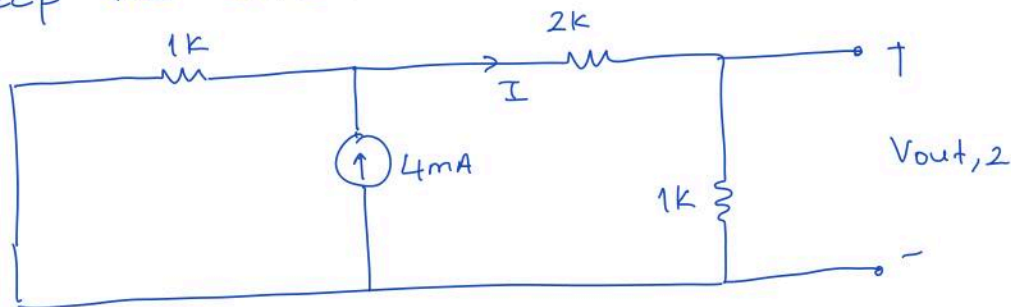


1) Keep the voltage source, kill other sources :



$$V_{out,1} = 10V \cdot \frac{1K}{1K + 2K + 1K} = 2.5V$$

2) Keep the current source, kill other sources

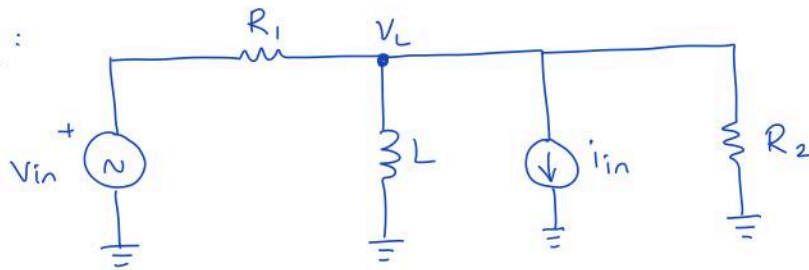


$$I = 4mA \cdot \frac{1K}{1K + 3K} = 1mA \quad (\text{current divider})$$

$$V_{out,2} = I \times 1K\Omega = 1V$$

3) Finally, add the results:

$$V_{out} = V_{out,1} + V_{out,2} = 3.5V$$

Example :

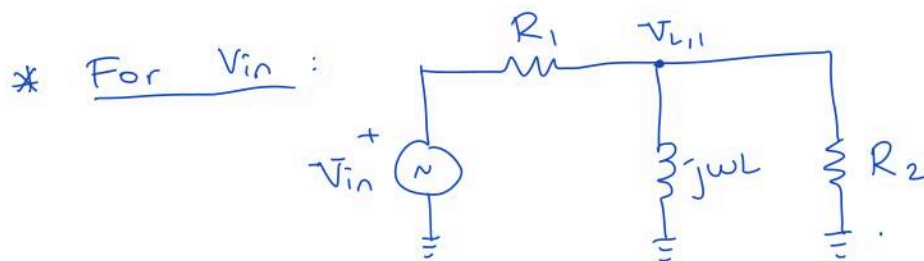
$$V_L(t) = ?$$

$$R_1 = 1\text{ k}\Omega, \quad R_2 = 3\text{ k}\Omega, \quad L = 0.5\text{ mH}$$

$$V_{in}(t) = 4\sqrt{2} \cos(15 \cdot 10^5 t + \frac{\pi}{4}) \text{ Volts}$$

$$i_{in}(t) = 8 \cos(\sqrt{3} \cdot 15 \cdot 10^5 t) \text{ mA}$$

Inputs are at different frequencies $\Rightarrow$ use superposition in time domain !



$$V_{in} = 4\sqrt{2} e^{j\frac{\pi}{4}} \text{ V}$$

$$Z_L = j\omega L = j \cdot 15 \cdot 10^5 \times 0.5 \times 10^{-3} \\ = j 750 \Omega = j 0.75 \text{ k}\Omega$$

$$V_{L,1} = V_{in} \cdot \frac{(j\omega L \parallel R_2)}{R_1 + (j\omega L \parallel R_2)} = V_{in} \cdot \frac{\frac{j\omega L \cdot R_2}{j\omega L + R_2}}{R_1 + \frac{j\omega L \cdot R_2}{j\omega L + R_2}}$$

$$= V_{in} \cdot \frac{(j\omega L)R_2}{R_1 R_2 + (j\omega L)(R_1 + R_2)}$$

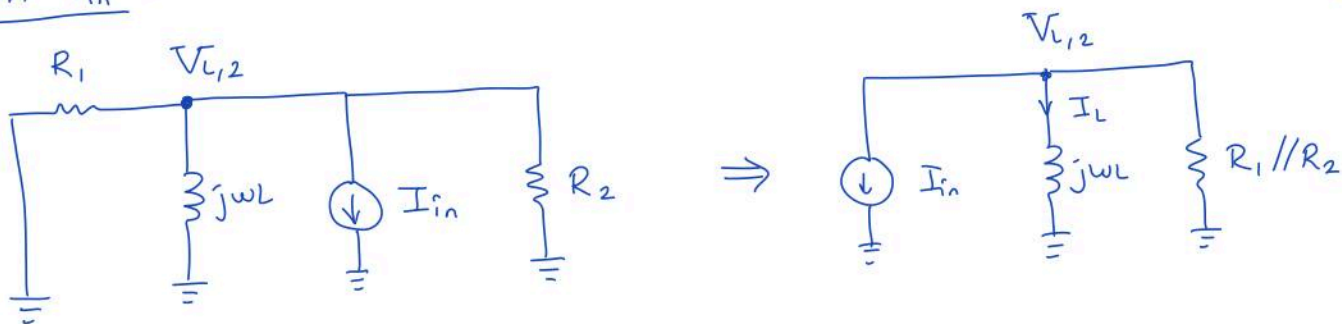
$$= 4\sqrt{2} \cdot e^{j\pi/4} \cdot \frac{j 0.75 \text{ k} \cdot 3 \text{ k}}{1 \text{ k} \cdot 3 \text{ k} + j 0.75 \text{ k} \cdot (4 \text{ k})}$$

$$= 4\sqrt{2} e^{j\pi/4} \frac{j 0.75 \times 3}{3 + j 0.75 \times 4}$$

$$= 4\sqrt{2} e^{j\pi/4} \frac{j 3}{4 + j 4} = 4\sqrt{2} \cdot e^{j\pi/4} \frac{3 \cdot e^{j\pi/2}}{4\sqrt{2} \cdot e^{j\pi/4}} = 3 e^{j\pi/2} \text{ V}$$

$$V_{L,1}(t) = 3 \cos(15 \cdot 10^5 t + \frac{\pi}{2}) \text{ V}$$

* From $\hat{I}_{in}$:



$$I_{in} = 8 \text{ mA}$$

$$\omega_2 = \sqrt{3} \cdot 15 \cdot 10^5 \text{ rps}$$

$$Z_L = jwL = j \sqrt{3} \cdot 15 \cdot 10^5 \times 0.5 \cdot 10^{-3} = j \sqrt{3} 750 \Omega = j \sqrt{3} 0.75 \text{ k}\Omega$$

$$I_L = -I_{in} \cdot \frac{R_1 // R_2}{jwL + R_1 // R_2} = -8 \times 10^{-3} \cdot \frac{0.75 \text{ k}}{j \sqrt{3} \cdot 0.75 \text{ k} + 0.75 \text{ k}}$$

$$= -8 \times 10^{-3} \cdot \frac{1}{1 + j\sqrt{3}} \text{ mA}$$

$$V_{L,2} = I_L \cdot jwL = -\frac{8 \cdot 10^{-3}}{1 + j\sqrt{3}} \cdot j \sqrt{3} \cdot 0.75 \text{ k}$$

$$= \frac{-j 6 \sqrt{3}}{1 + j\sqrt{3}} = \frac{6 \sqrt{3} e^{-j\pi/2}}{2 e^{j\pi/3}} = 3 \sqrt{3} e^{-j5\pi/6} \text{ V}$$

Note: A phasor diagram shows a vector I_{in} on the positive real axis and a vector I_L at an angle of $-5\pi/6$ from the positive real axis.

$$V_{L,2}(t) = 3\sqrt{3} \cos(\sqrt{3} \cdot 15 \cdot 10^5 t - \frac{5\pi}{6}) \text{ V}$$

* Now, add the results in time domain:

$$\begin{aligned} V_L(t) &= V_{L,1}(t) + V_{L,2}(t) \\ &= 3 \cos(15 \cdot 10^5 t + \frac{\pi}{2}) + 3\sqrt{3} \cos(\sqrt{3} \cdot 15 \cdot 10^5 t - \frac{5\pi}{6}) \text{ V} \end{aligned}$$